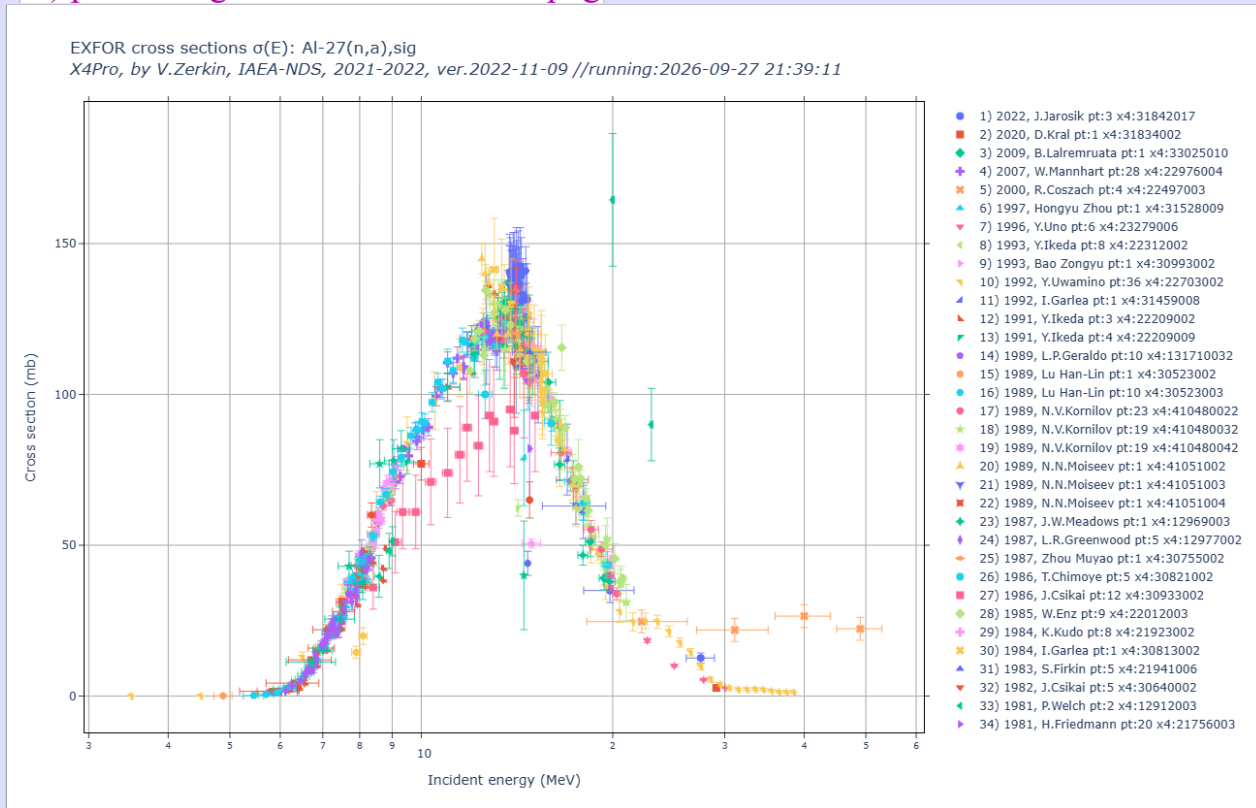
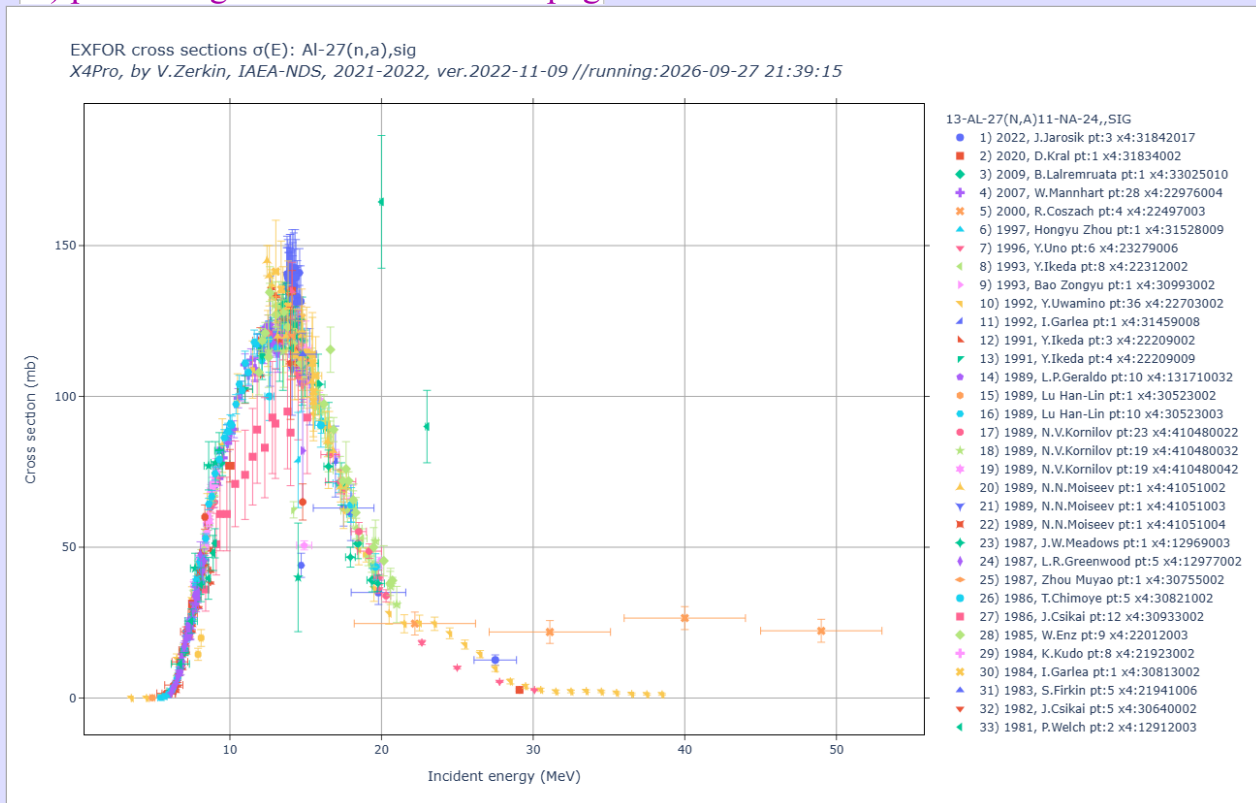


X4Pro. Plotting via Plotly.

Generated: 2026-09-28,13:49:32

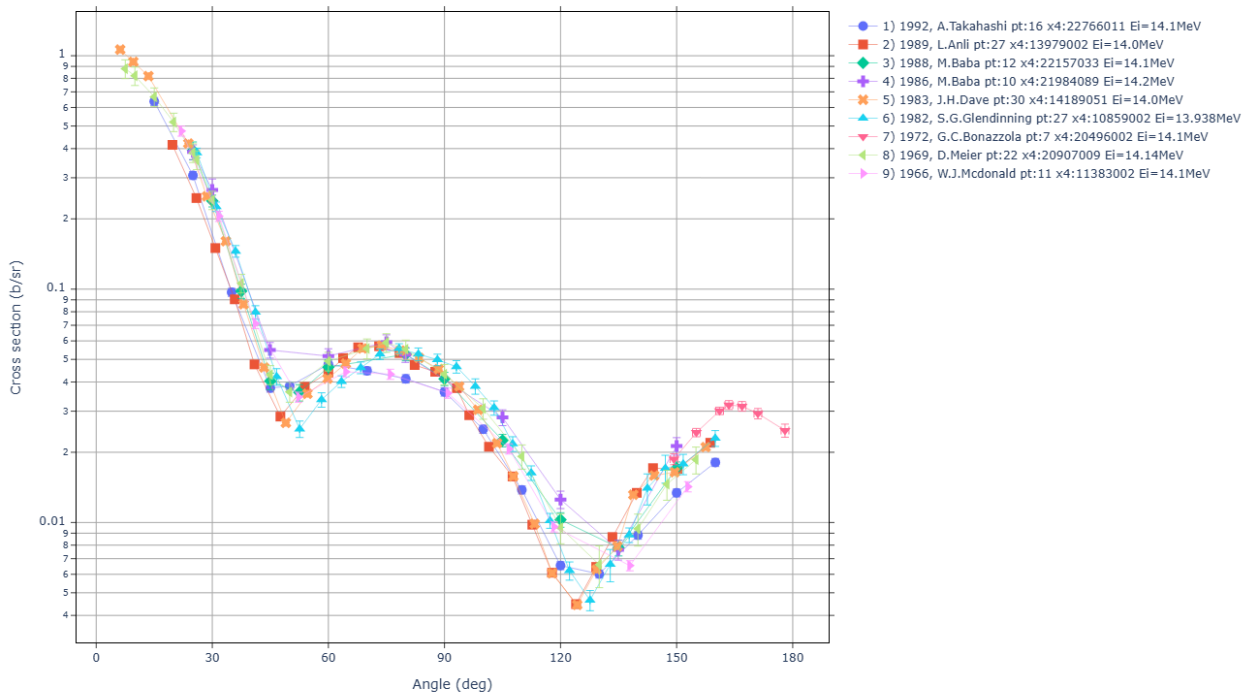
part1-1-sig/ Retrieve and plot EXFOR cross sections (modular Python-program)

1) part1-1-sig/out00/Al27na-1.html.png**2) part1-1-sig/out00/Al27na-2.html.png**

part1-2-da/ EXFOR angular distributions

3) part1-2-da/out00/da0an.html.png

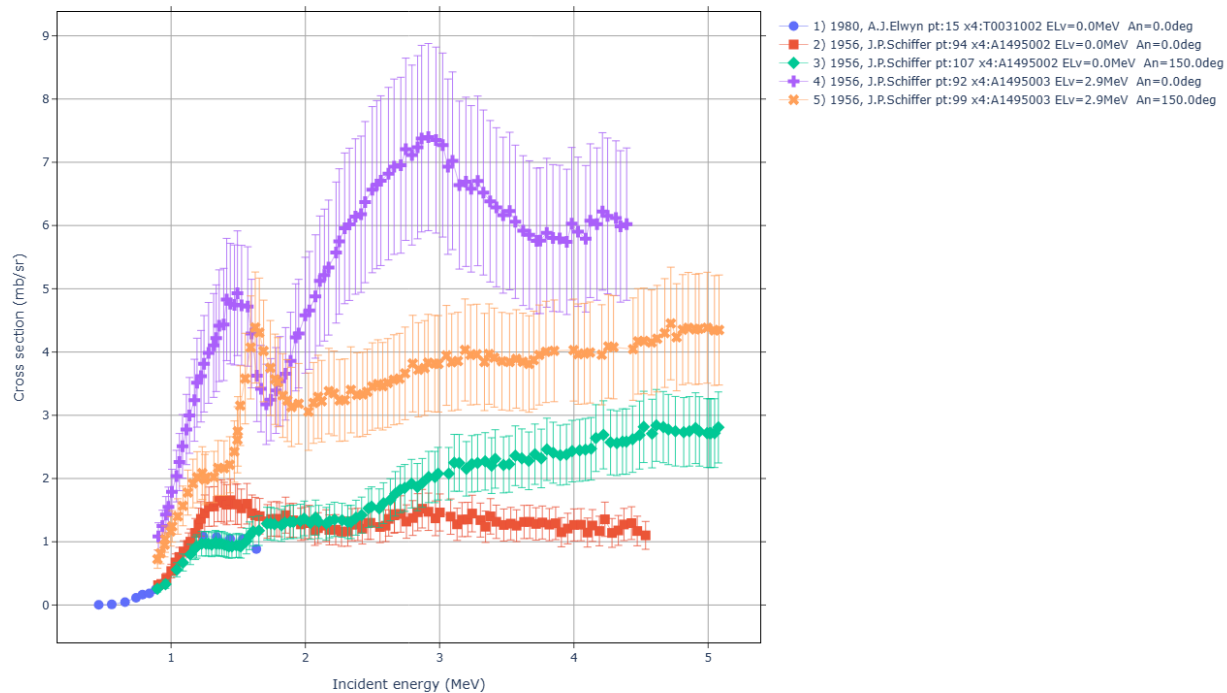
Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da
 X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-12-20 //running:2026-09-27 21:39:18



part1-3-dap/ EXFOR partial angular distributions

4) part1-3-dap/out00/dap0ei.html.png

Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: Li-6(he3,p)par,da
 X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:39:20

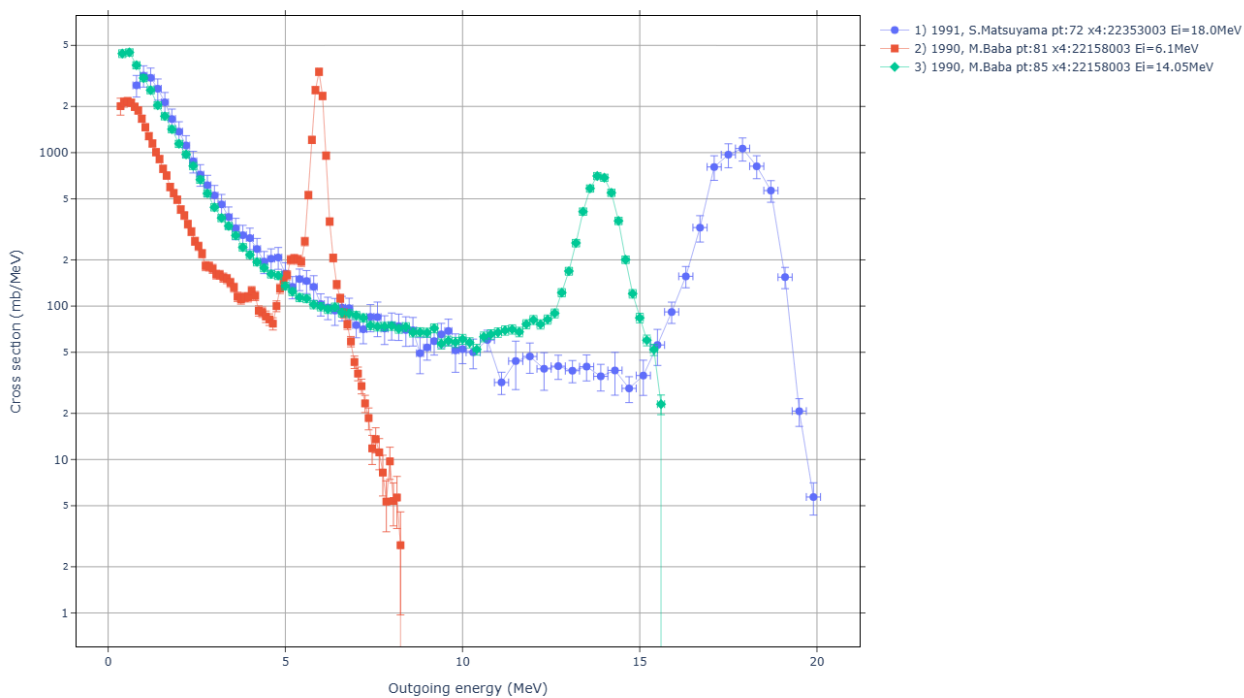


part1-4-de/ EXFOR energy spectra of outgoing particles

5) part1-4-de/out00/de0eo.html.png

Plot EXFOR emission spectra $d\sigma/dE_{out}$: Th-232(n,xn),de

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-12-20 //running:2026-09-27 21:39:23

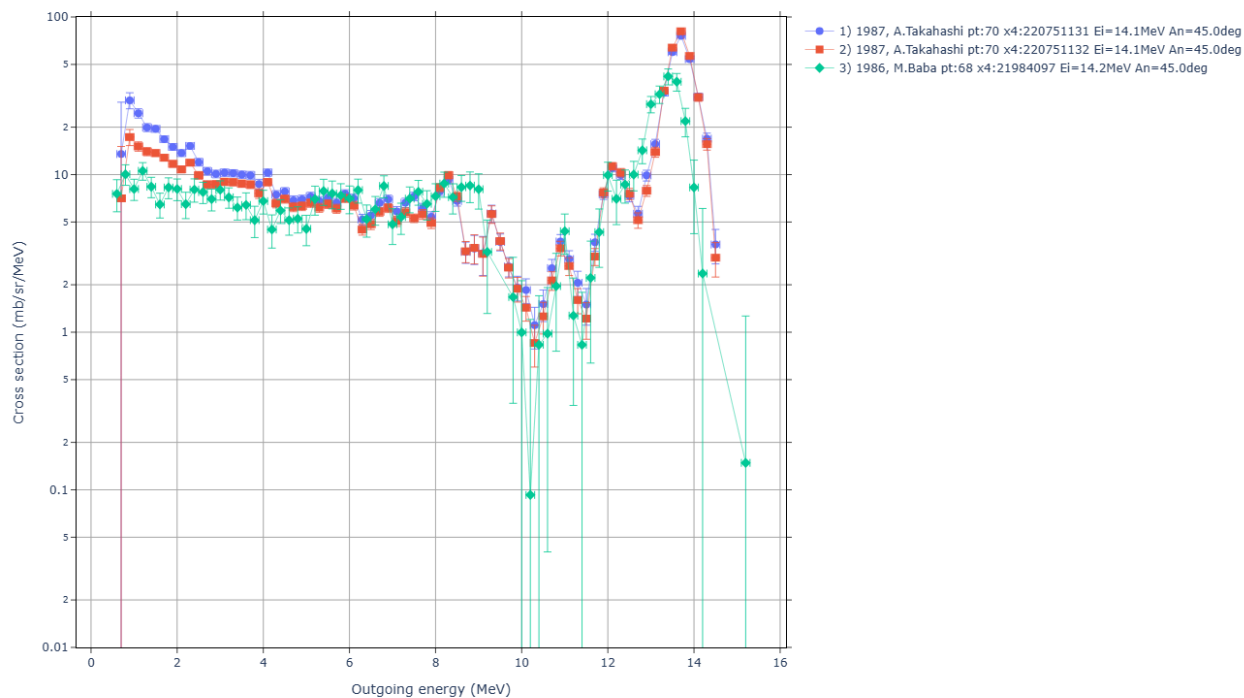


part1-5-dae/ EXFOR double differential cross sections

6) part1-5-dae/out00/dae0eo.html.png

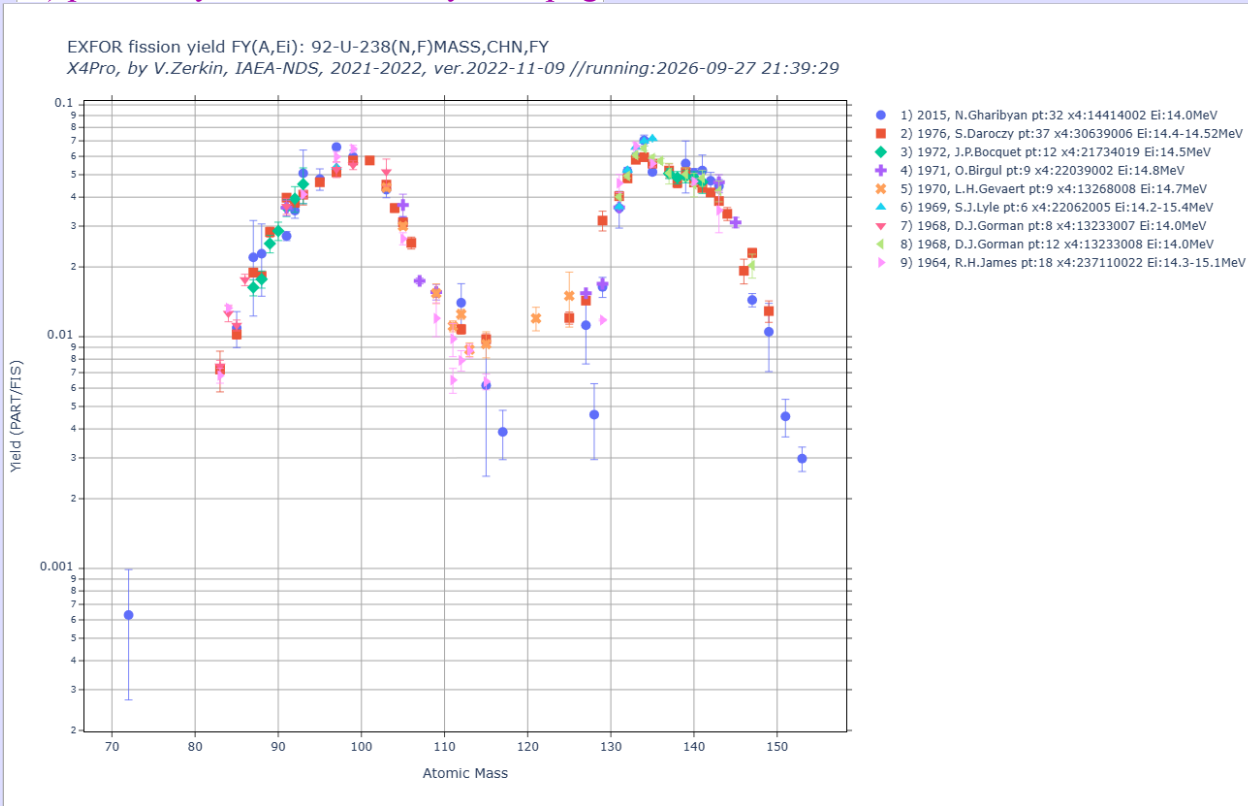
Plot EXFOR double differential cross sections: F-19(n,xn),dae

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:39:26



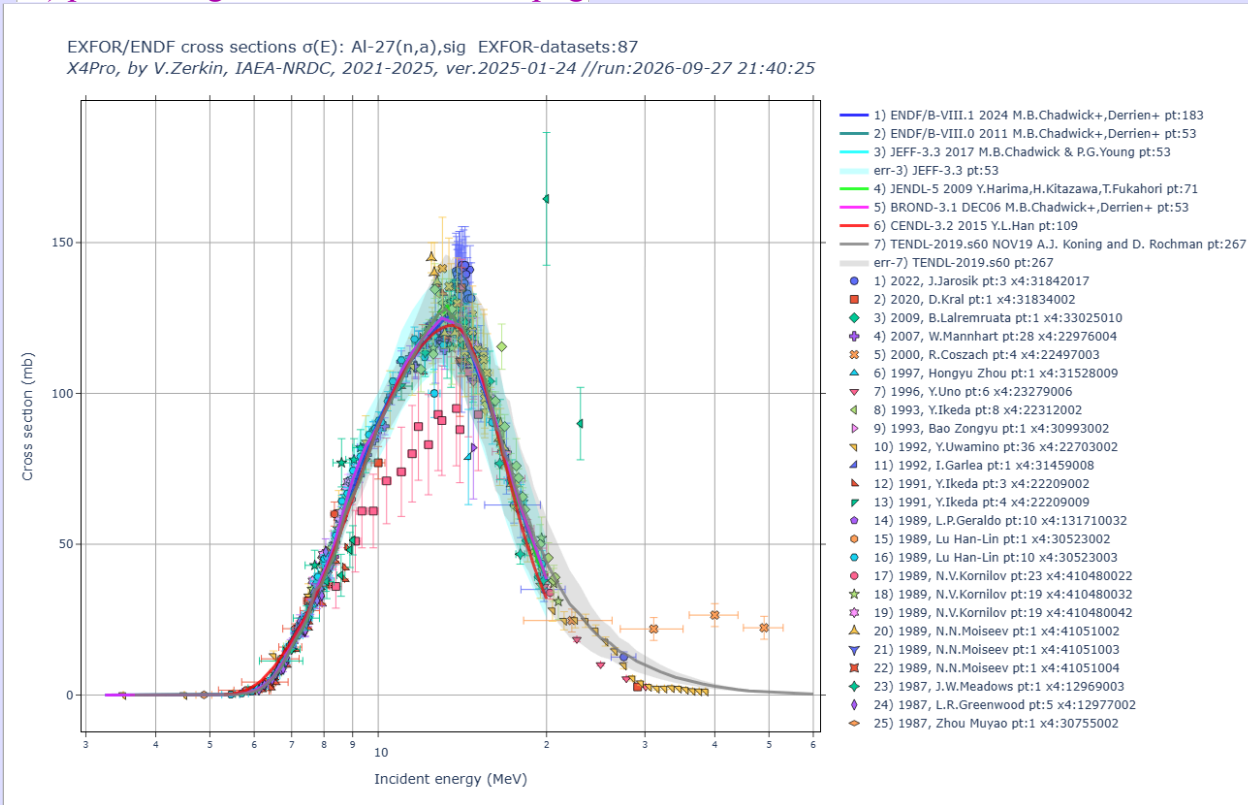
part1-6-fy/ EXFOR fission yield

7) part1-6-fy/out00/U238nf-fy.html.png



part2-1-sig1/ Retrieve and plot EXFOR/ENDF cross sections

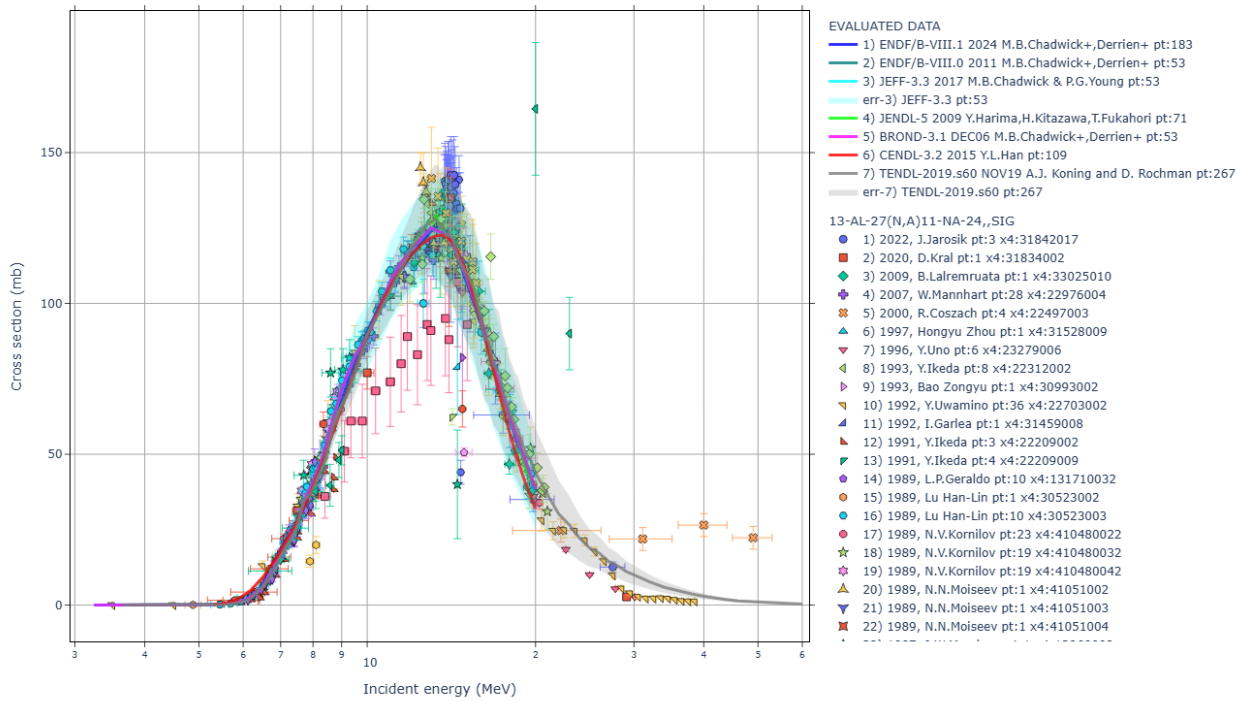
8) part2-1-sig1/out00/Al27na.html.png



9) part2-1-sig1/out00/Al27na2.html.png

EXFOR/ENDF cross sections $\sigma(E)$: Al-27(n,a),sig EXFOR-datasets:93

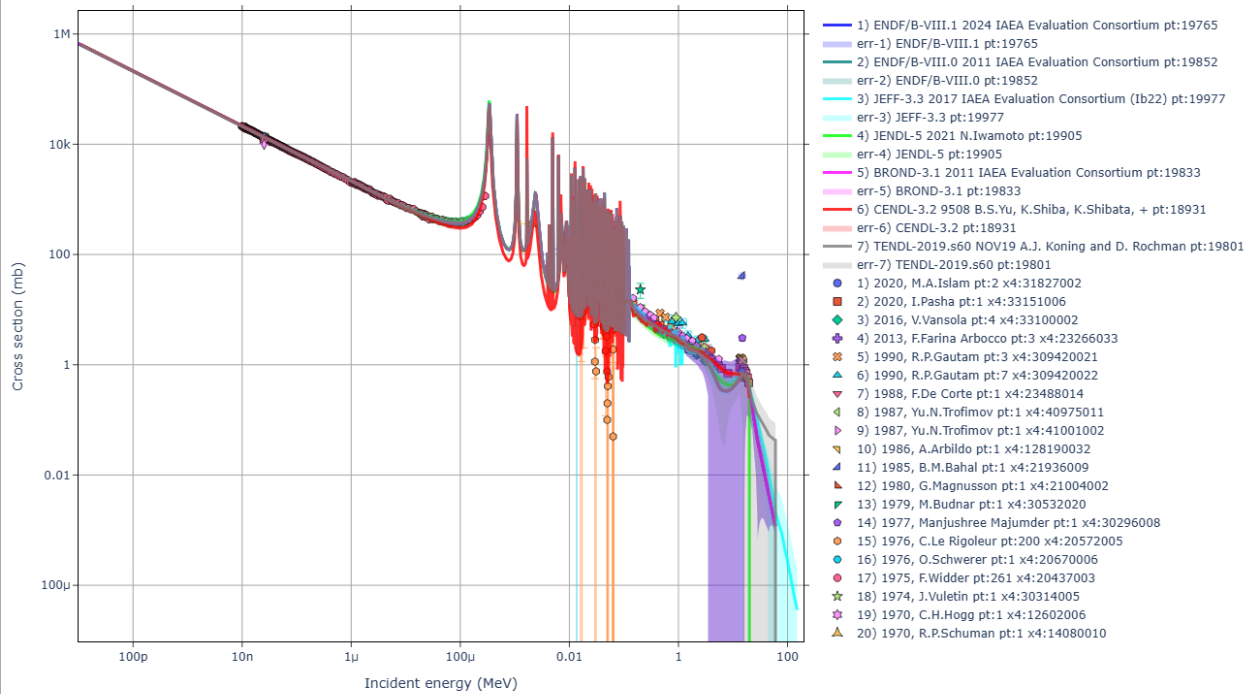
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-09-27 21:41:23



10) part2-1-sig1/out00/mn55ng.html.png

EXFOR/ENDF cross sections $\sigma(E)$: mn-55(n,g),sig EXFOR-datasets:49

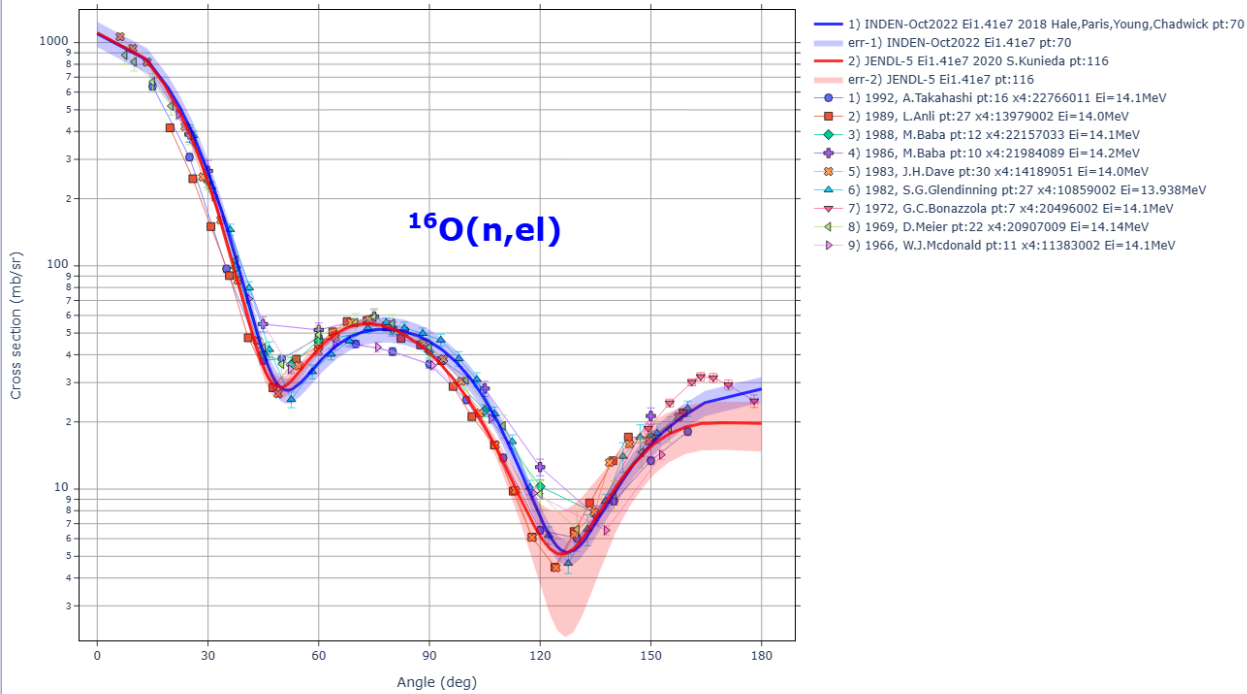
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-09-27 21:41:26



part2-2-da1an/ EXFOR/ENDF angular distributions (Ei:fixed)

11) part2-2-da1an/out00/da1an.html.png

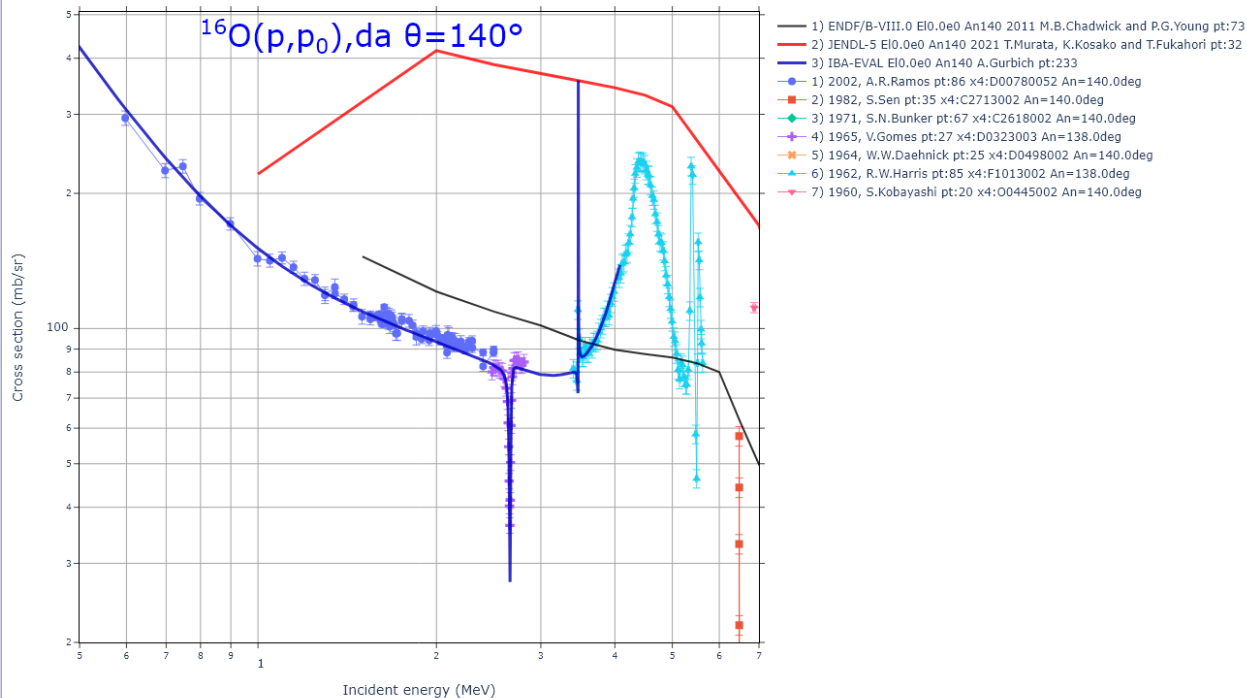
EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da EXFOR-datasets:9
 X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-09-27 21:42:25



part2-3-da1ei/ EXFOR/ENDF angular distributions (angle:fixed)

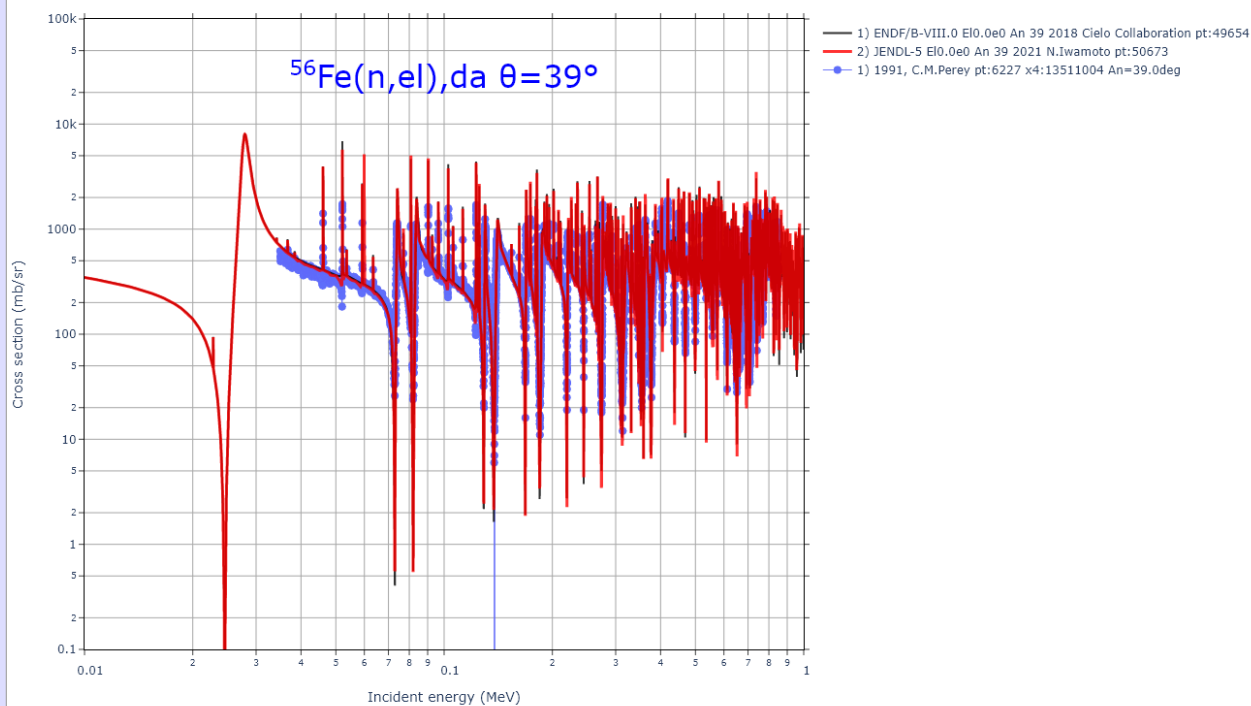
12) part2-3-da1ei/out00/da1ei-ex1.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(p,p₀),da $\theta=140^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-09-28 00:53:18



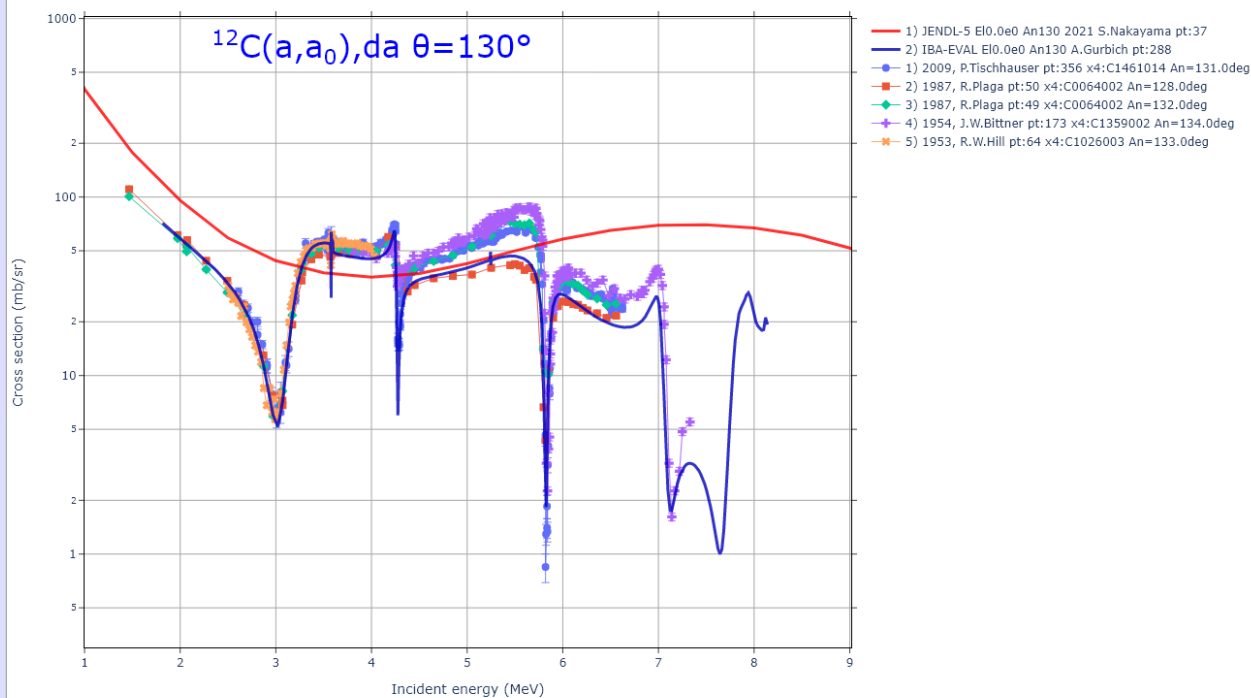
13) part2-3-da1ei/out00/da1ei-ex2.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: Fe-56(n,el), da $\theta=39^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-09-28 00:53:20



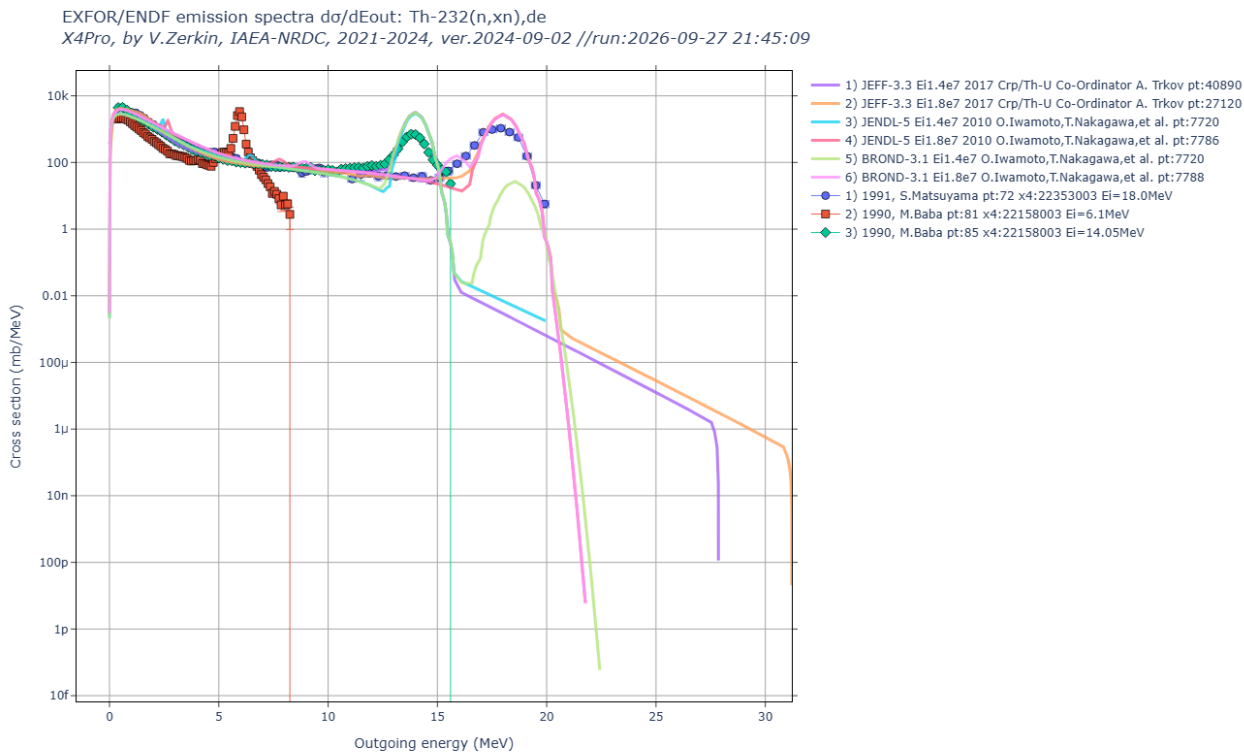
14) part2-3-da1ei/out00/da1ei.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: C-12(a,el), da $\theta=130^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-09-28 00:53:13



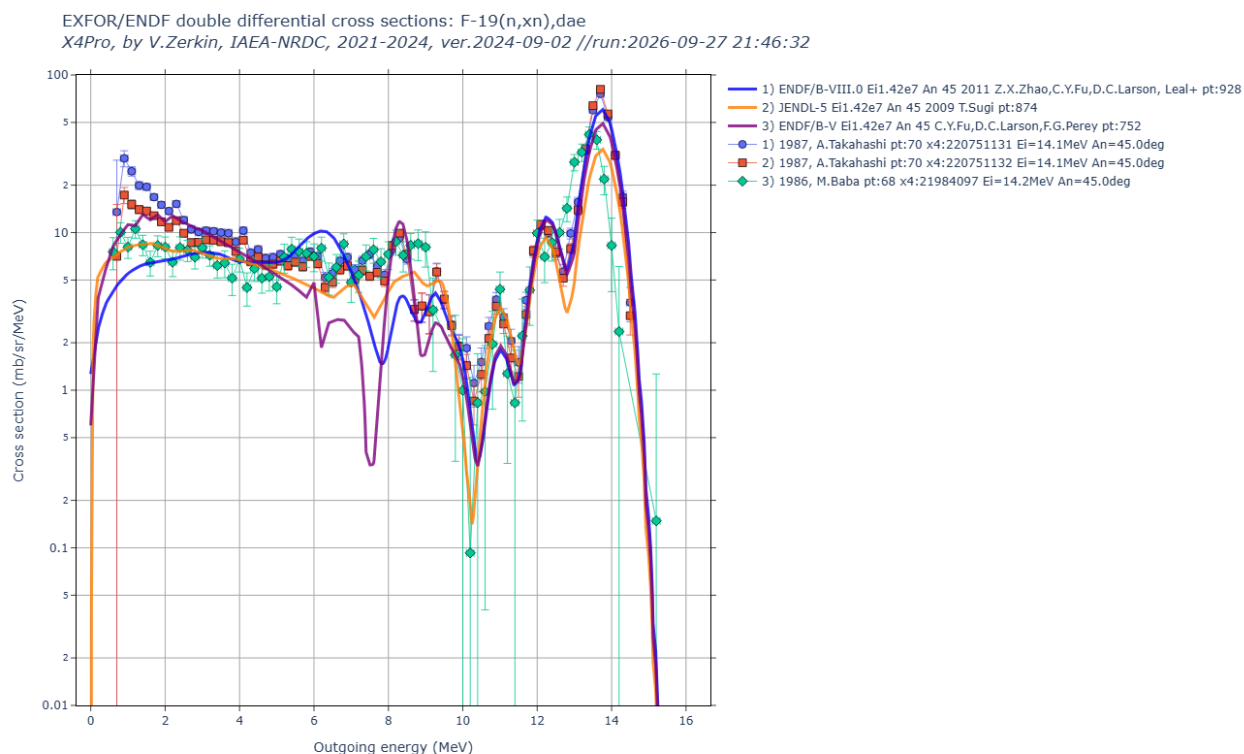
part2-4-de1eo/ EXFOR/ENDF energy spectra of outgoing particles

15) part2-4-de1eo/out00/de1eo.html.png



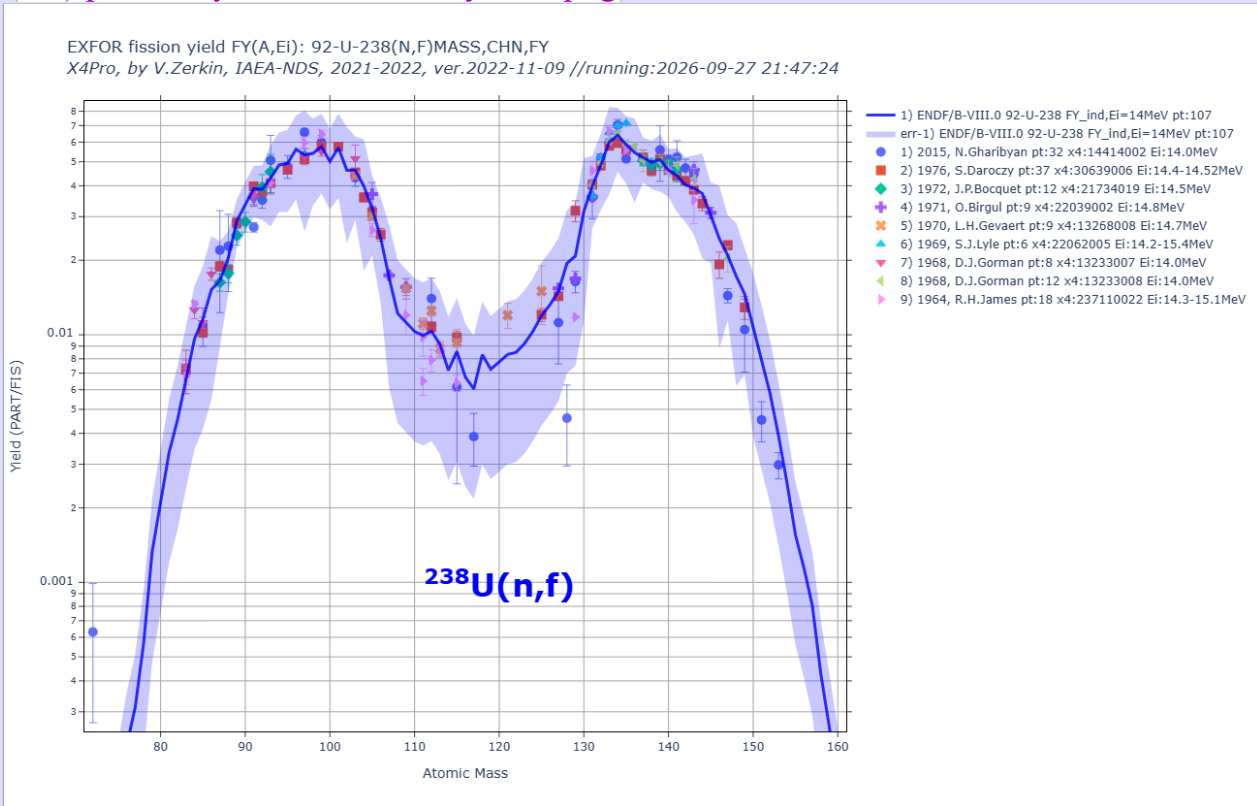
part2-5-dae1eo/ EXFOR/ENDF double differential cross sections

16) part2-5-dae1eo/out00/dae1eo.html.png



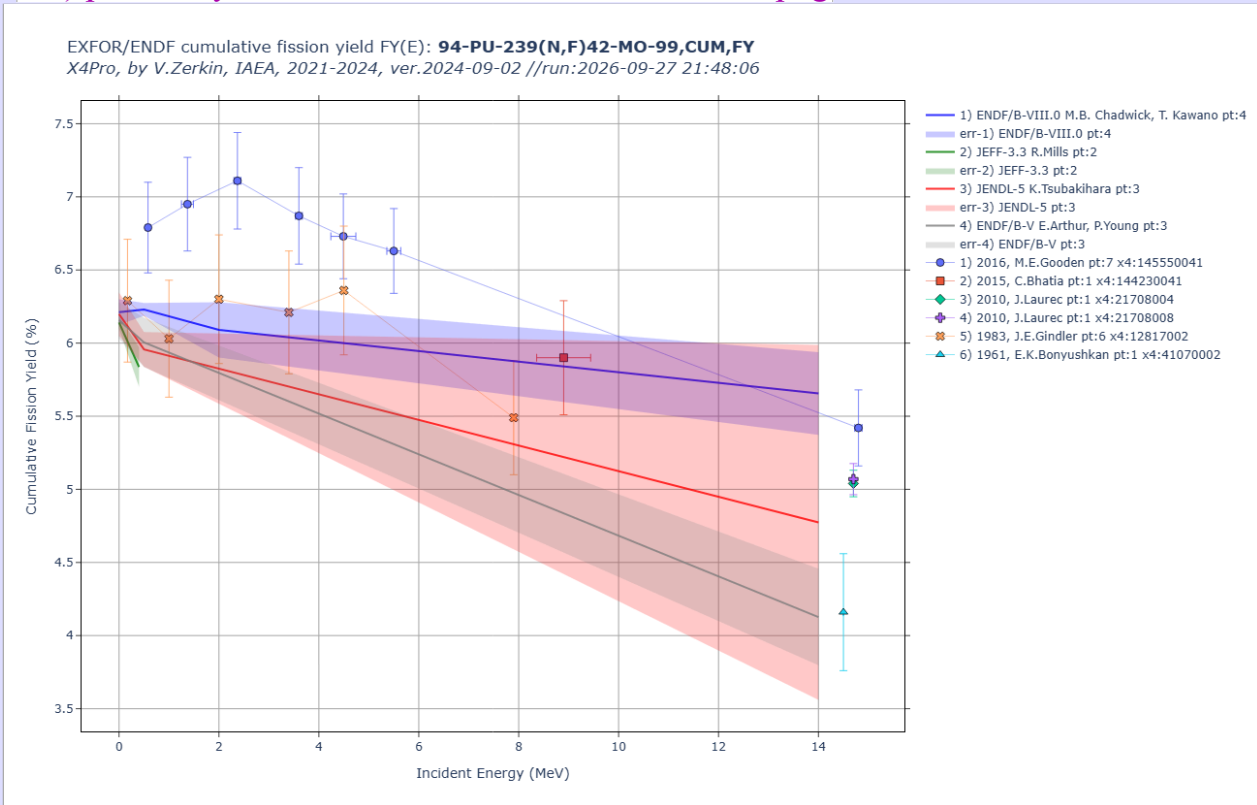
part2-6-fy/ EXFOR/ENDF fission yeild (mass distribution)

17) part2-6-fy/out00/U238nf-fy.html.png



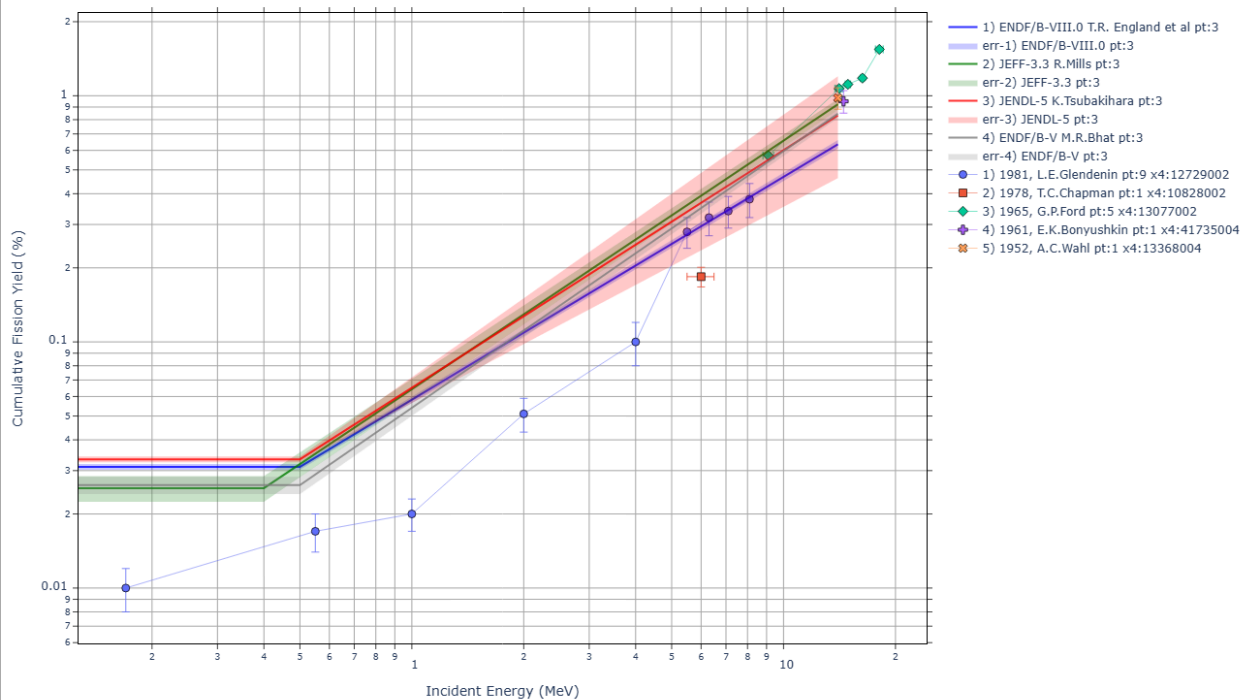
part2-7-fye/ EXFOR/ENDF fission yeild (energy distribution)

18) part2-7-fye/out00/Pu239-nf-Mo99-cumFY.html.png



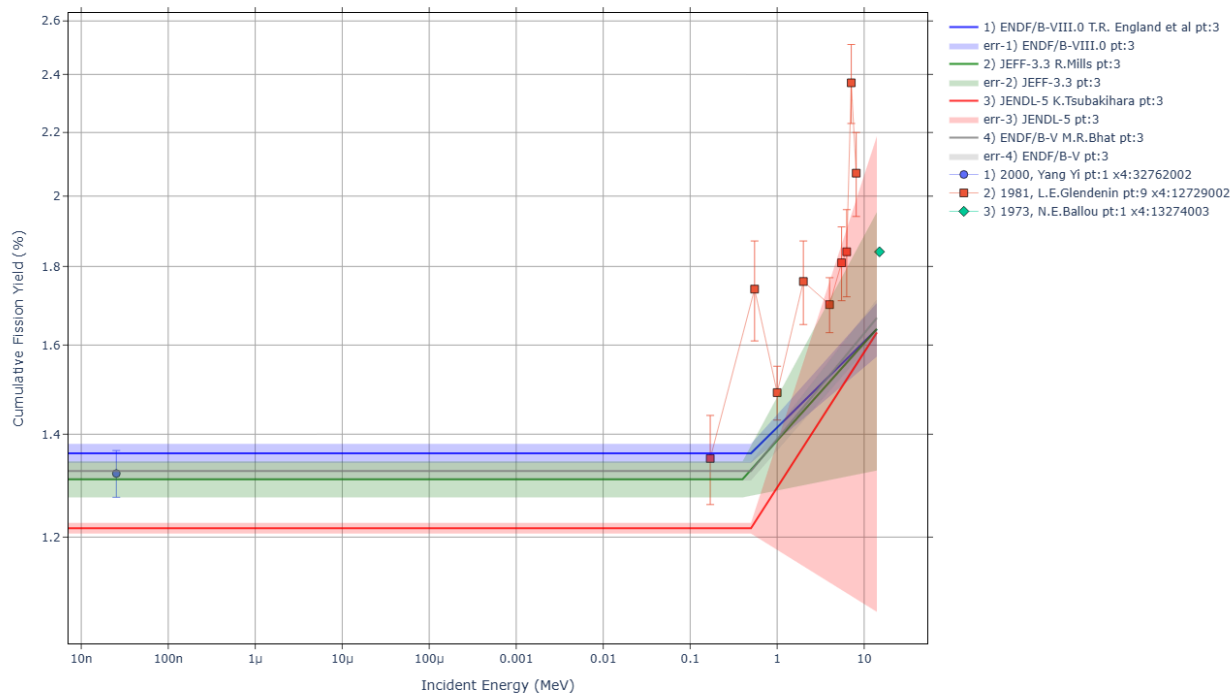
19) part2-7-fye/out00/u235-nf-cd115g-cumFY.html.png

EXFOR/ENDF cumulative fission yield FY(E): **92-U-235(N,F)48-CD-115-G,CUM,FY**
 X4Pro, by V.Zerkin, IAEA, 2021-2024, ver.2024-09-02 //run:2026-09-27 21:49:55



20) part2-7-fye/out00/u235-nf-kr85m-cumFY.html.png

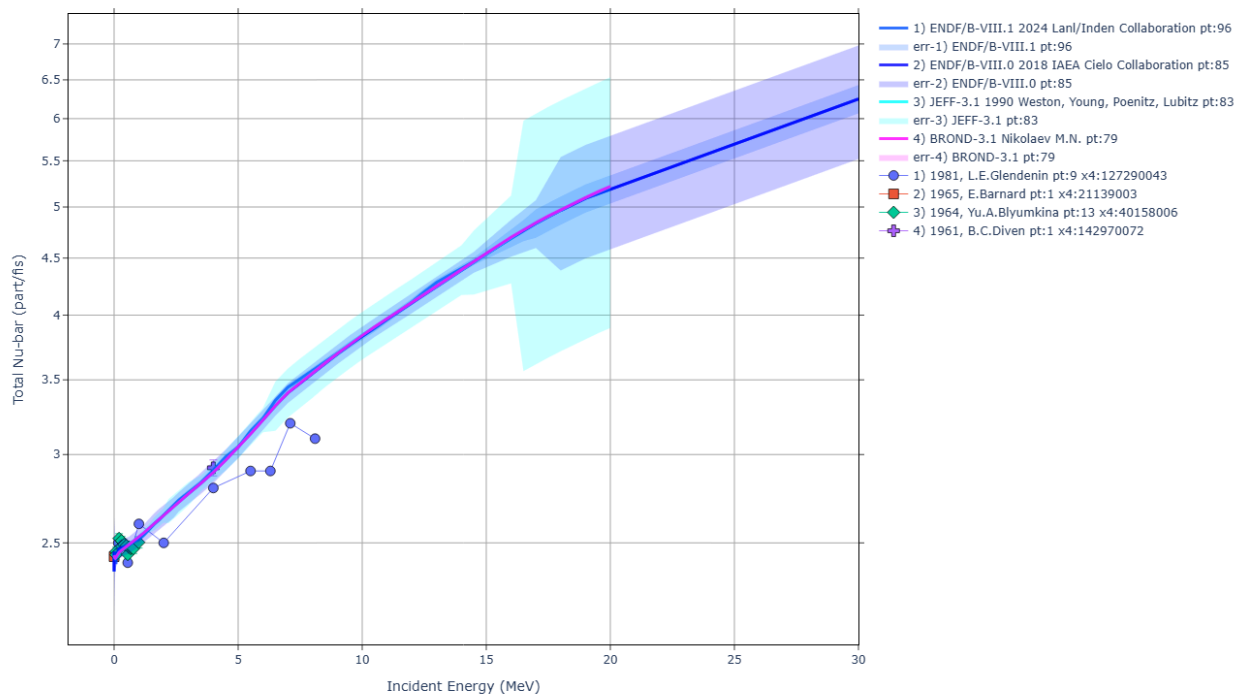
EXFOR/ENDF cumulative fission yield FY(E): **92-U-235(N,F)36-KR-85-M1,CUM,FY**
 X4Pro, by V.Zerkin, IAEA, 2021-2024, ver.2024-09-02 //run:2026-09-27 21:49:08



part2-9-nubar/ EXFOR/ENDF NUBAR - average number of neutrons per fission

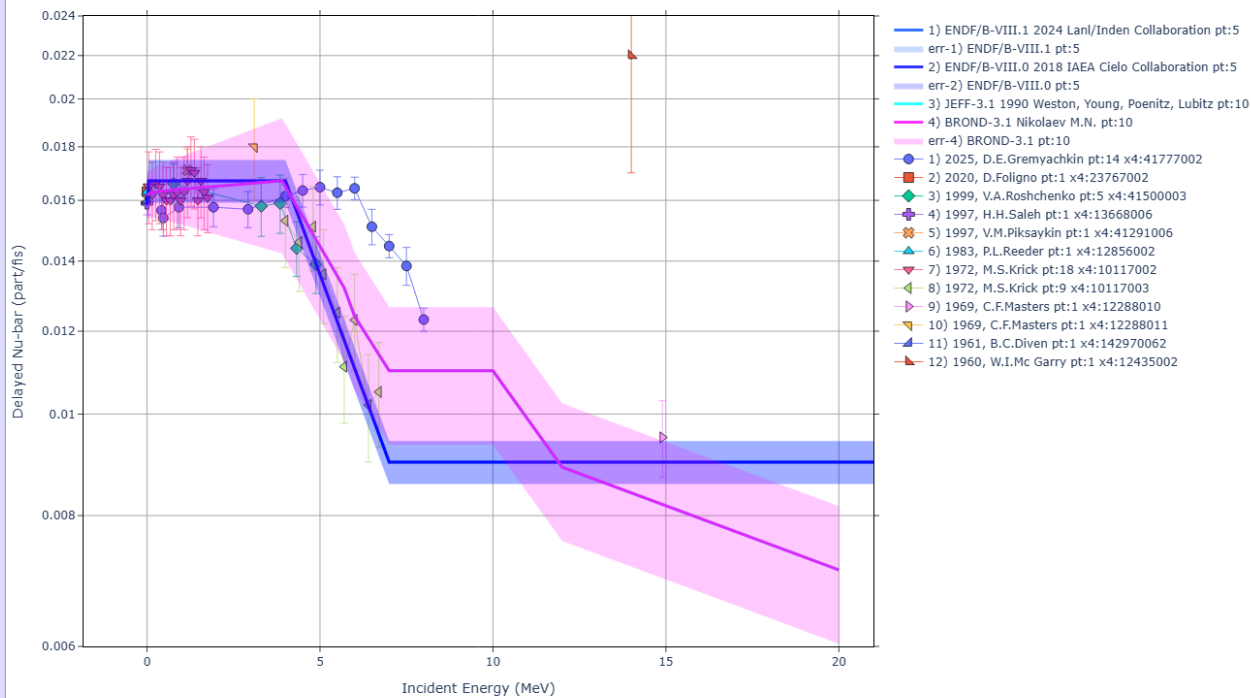
21) part2-9-nubar/out00/U235nf_-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),,NU Exp.Datasets:4 Points:24
 X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:51:54



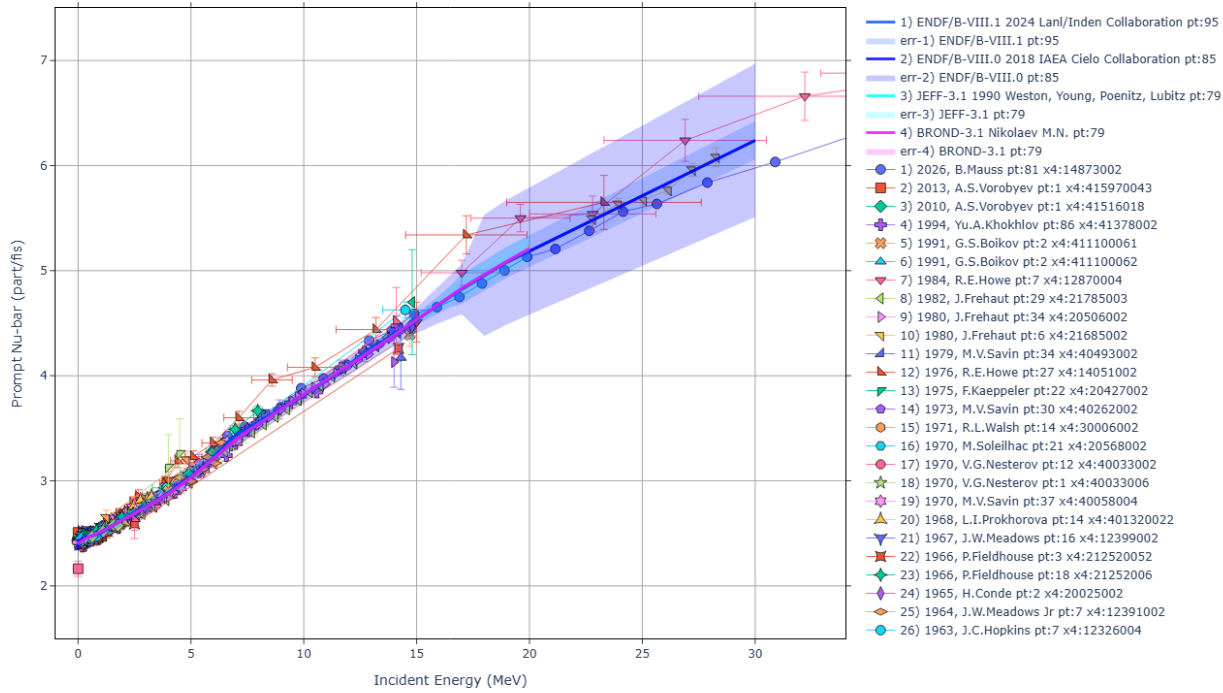
22) part2-9-nubar/out00/U235nf_dl-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),DL,NU Exp.Datasets:12 Points:54
 X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:51:38



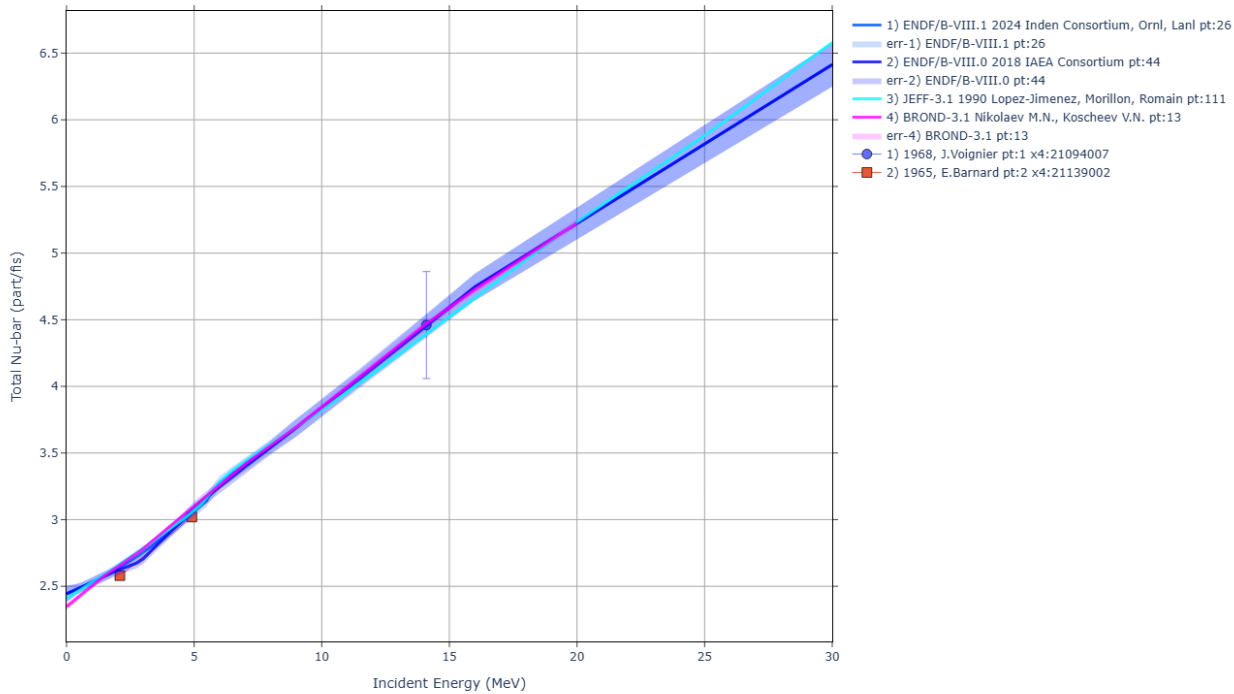
23) part2-9-nubar/out00/U235nf_pr-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),PR,NU Exp.Datasets:38 Points:535
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:50:46

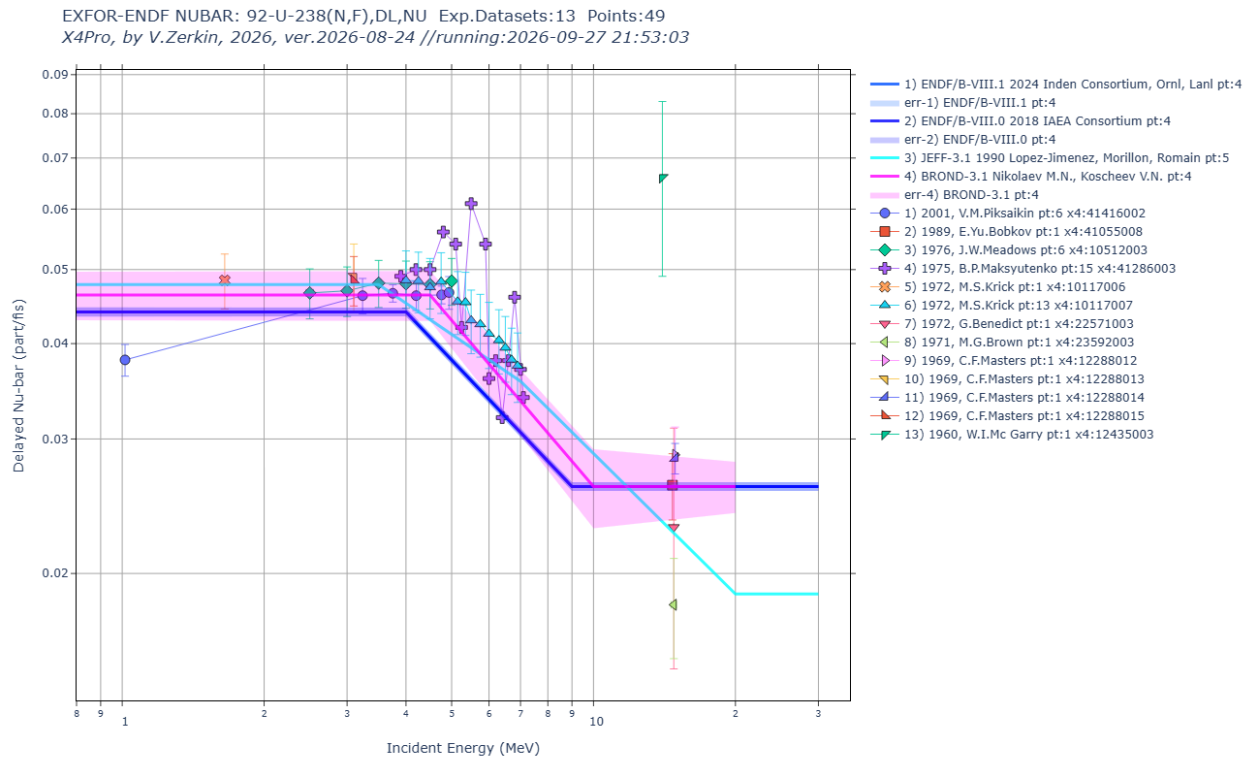


24) part2-9-nubar/out00/U238nf_-nu.html.png

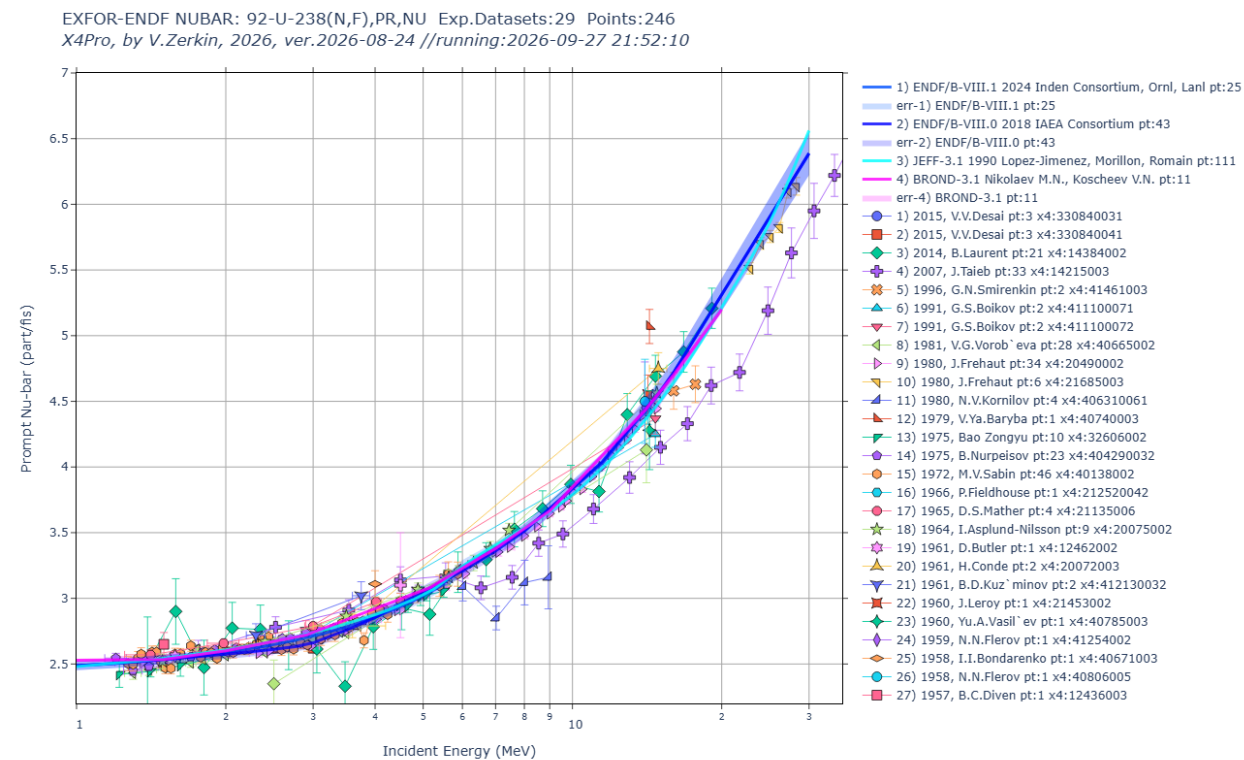
EXFOR-ENDF NUBAR: 92-U-238(N,F),,NU Exp.Datasets:2 Points:3
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:53:19



25) part2-9-nubar/out00/U238nf_dl-nu.html.png

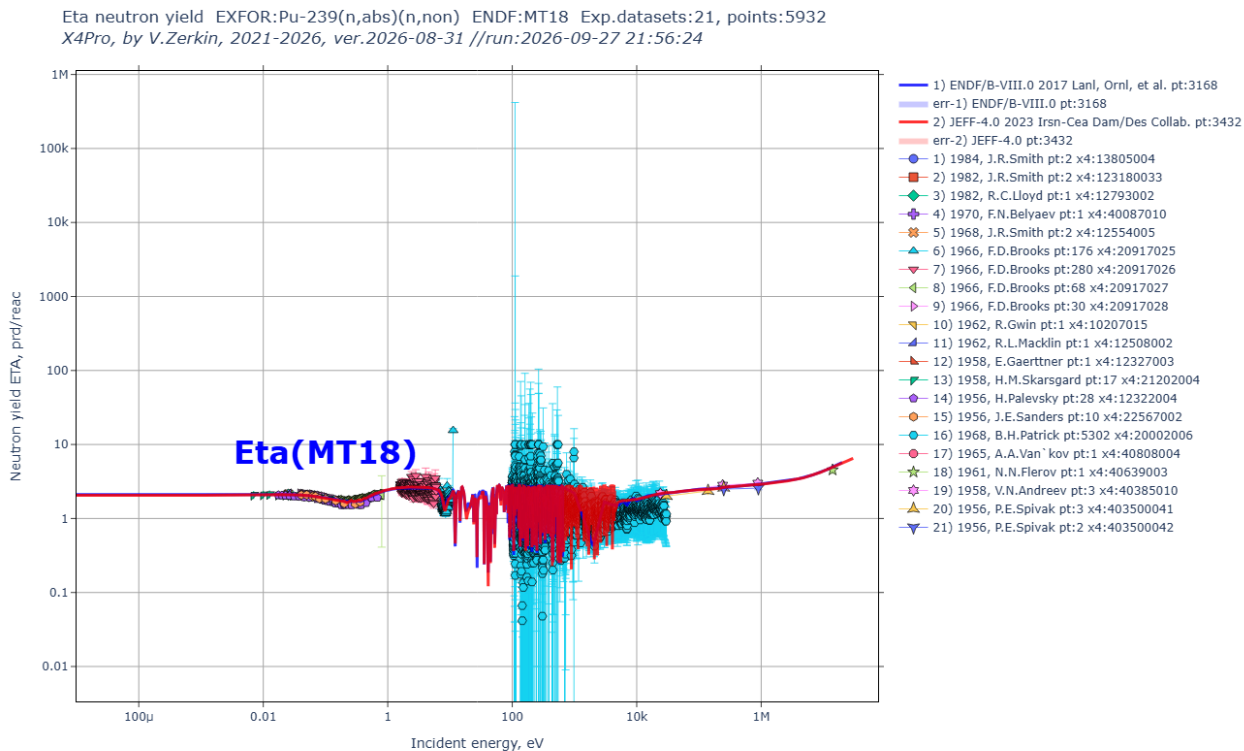


26) part2-9-nubar/out00/U238nf_pr-nu.html.png

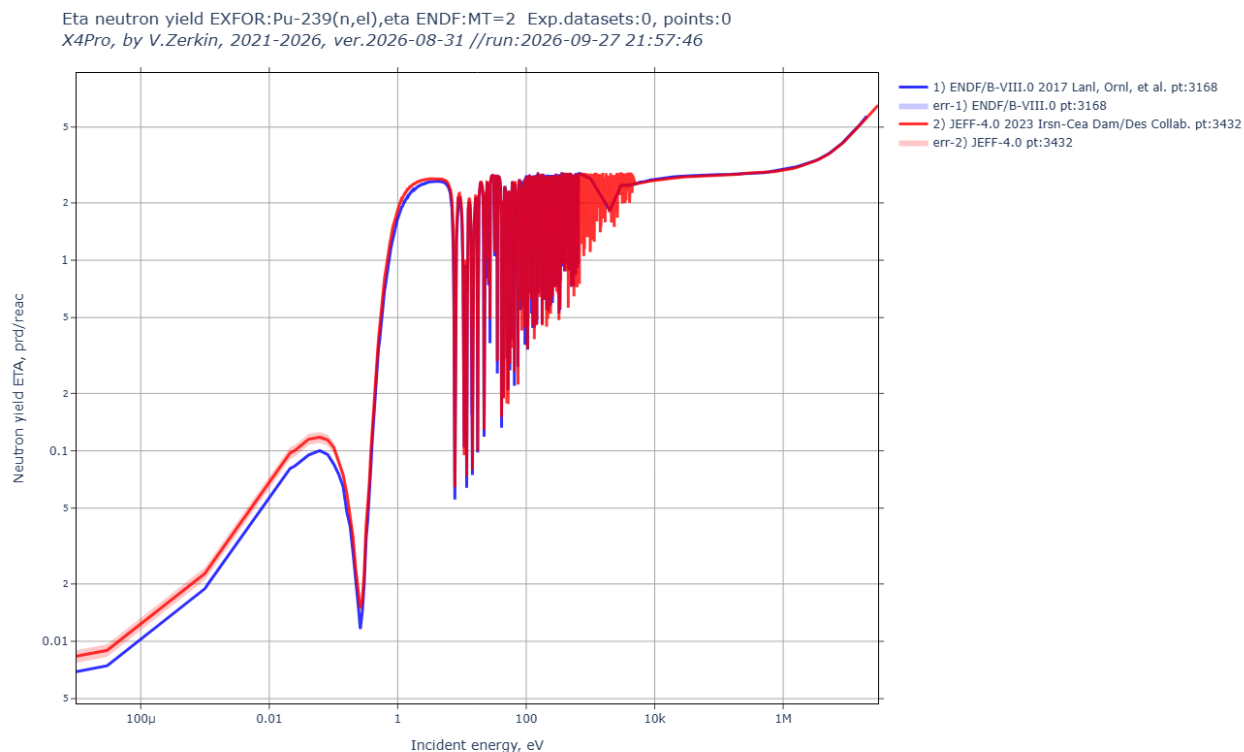


part2-A-eta/ EXFOR/ENDF Eta - average neutron yield per nonelastic event

27) part2-A-eta/out00/Pu239-eta.html.png

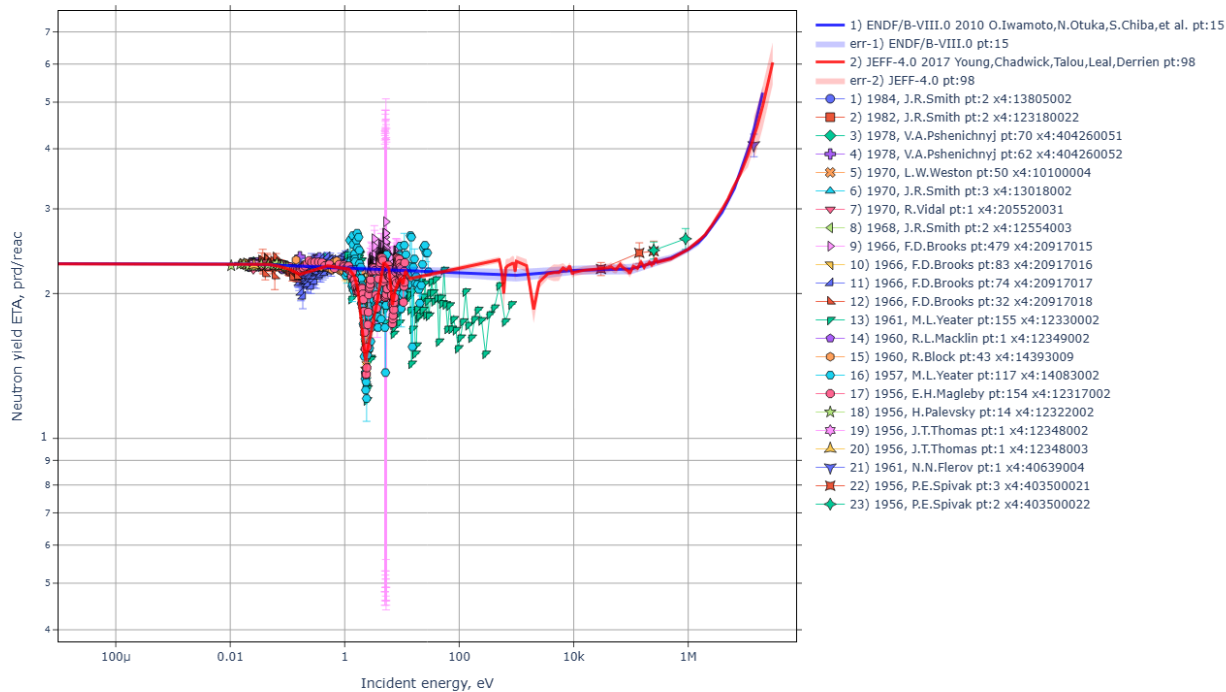


28) part2-A-eta/out00/Pu239nel-eta.html.png



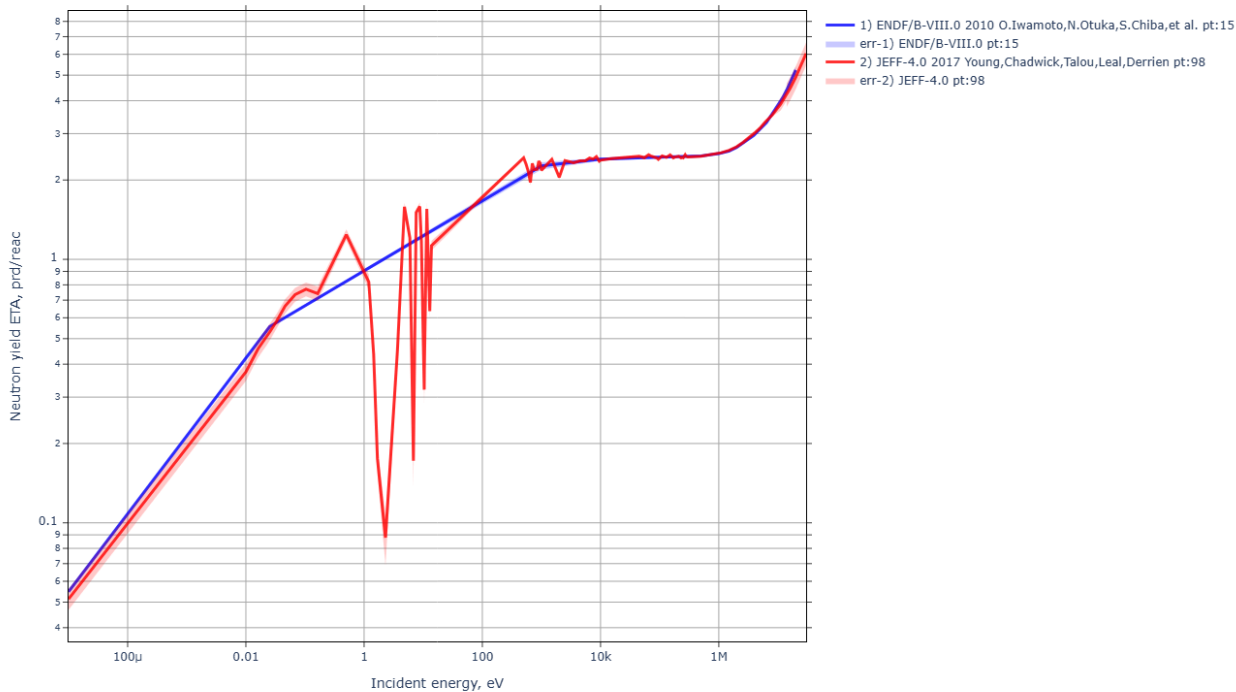
29) part2-A-eta/out00/U233-eta.html.png

Eta neutron yield EXFOR:U-233(n,abs)(n,non) ENDF:MT18 Exp.datasets:23, points:1352
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:55:37



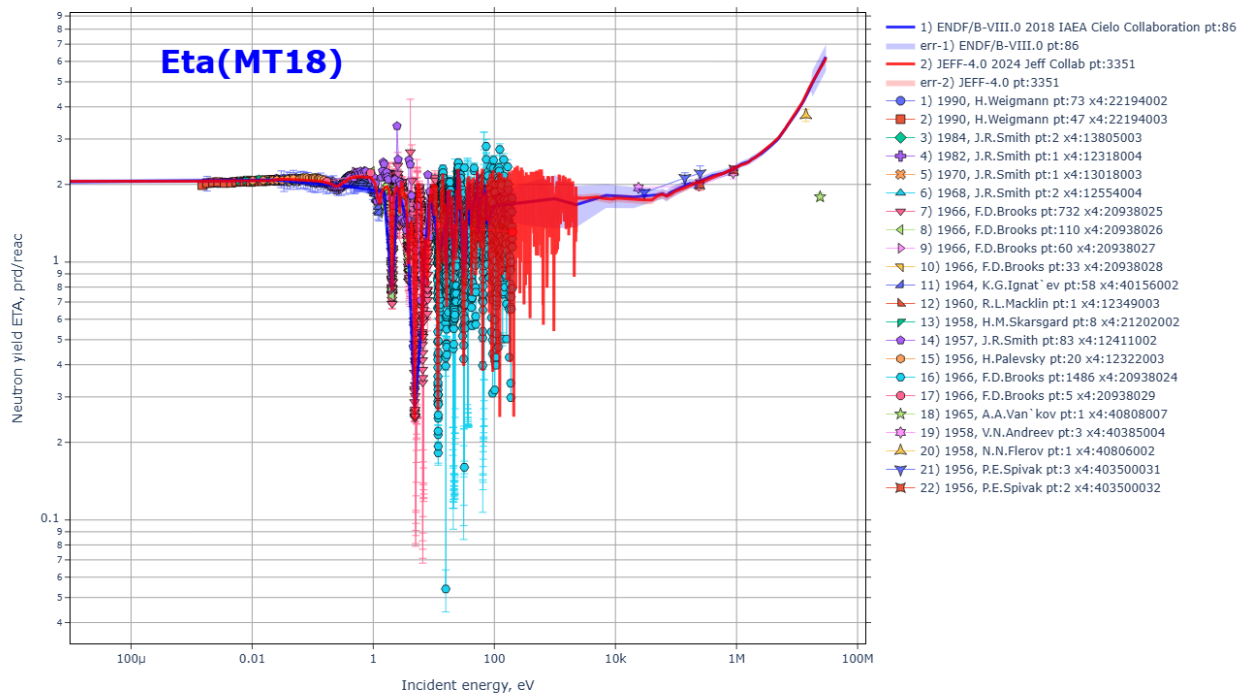
30) part2-A-eta/out00/U233nel-eta.html.png

Eta neutron yield EXFOR:U-233(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:57:36



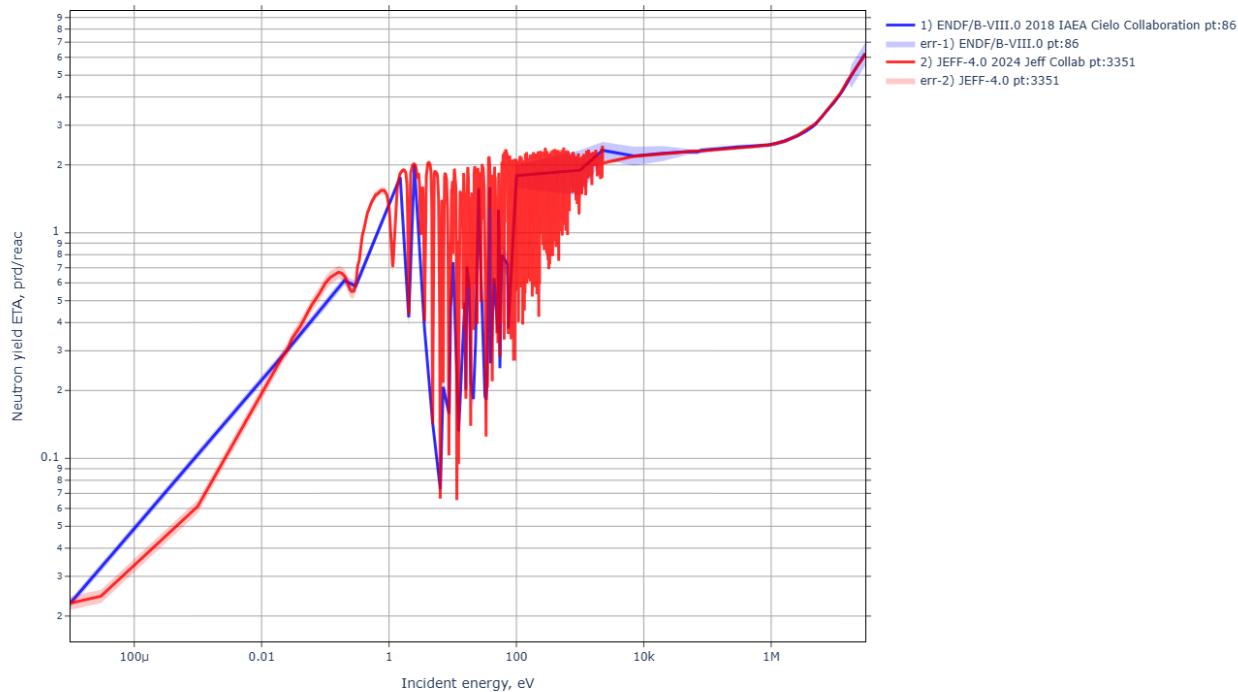
31) part2-A-eta/out00/U235-eta.html.png

Eta neutron yield EXFOR:U-235(n,abs)(n,non) ENDF:MT18 Exp.datasets:22, points:2732
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:53:59



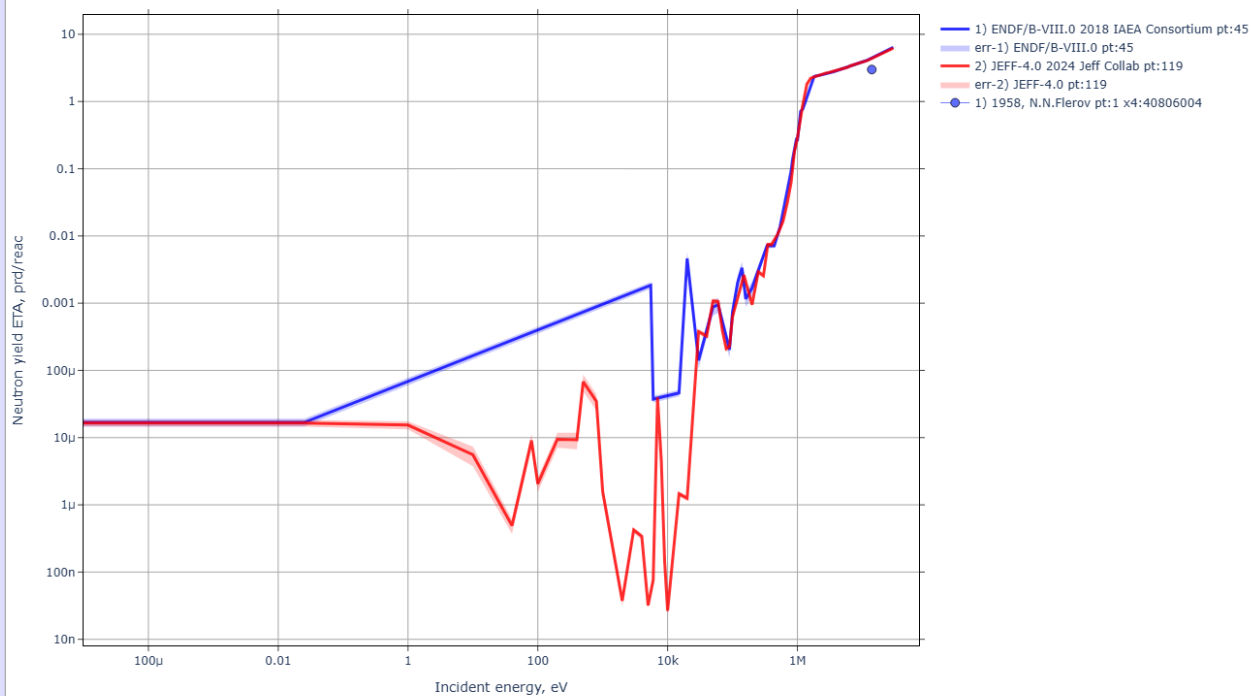
32) part2-A-eta/out00/U235nel-eta.html.png

Eta neutron yield EXFOR:U-235(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:57:10



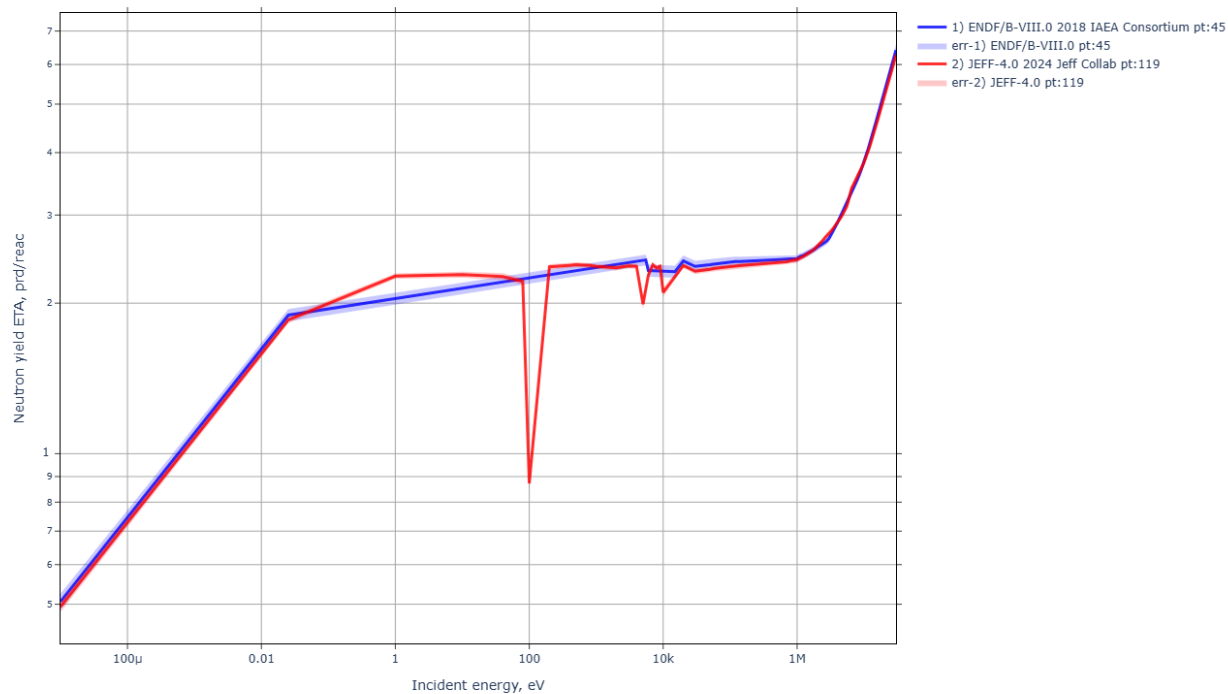
33) part2-A-eta/out00/U238-eta.html.png

Eta neutron yield EXFOR:U-238(n,non) ENDF:MT18 Exp.datasets:1, points:1
 X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:54:49



34) part2-A-eta/out00/U238nel-eta.html.png

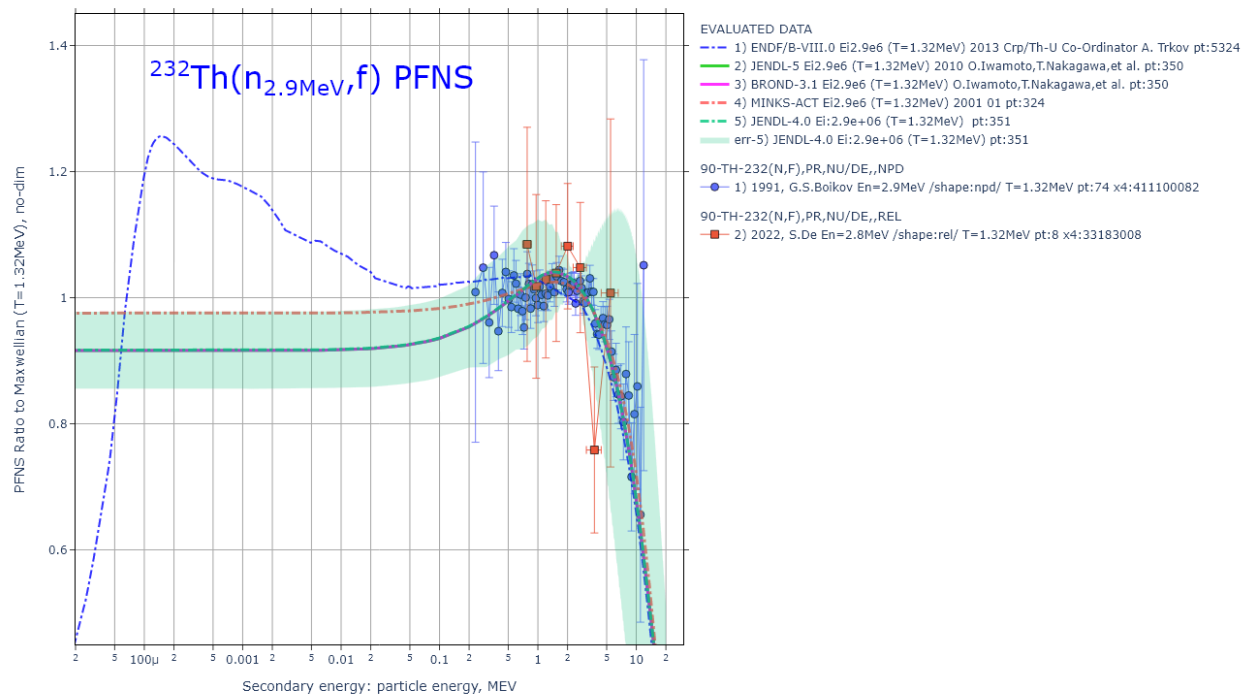
Eta neutron yield EXFOR:U-238(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
 X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:57:23



part2-B-pfns/ PFNS - energy spectrum of prompt fission neutrons

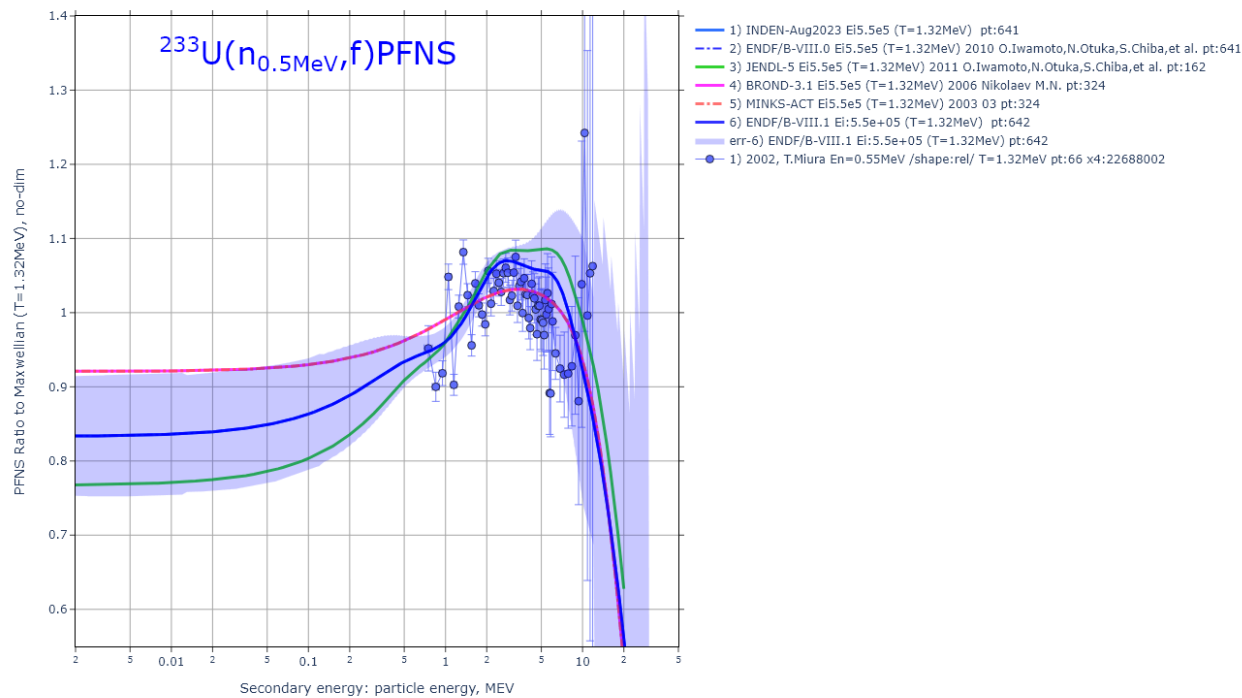
35) part2-B-pfns/out00/pfns232th2_6.html.png

Reaction: 90-TH-232(N,F),PR,NU/DE,,NPD Quantity: MFQ: y=Data(En,E2) Datasets: 2 Points: 82
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:43



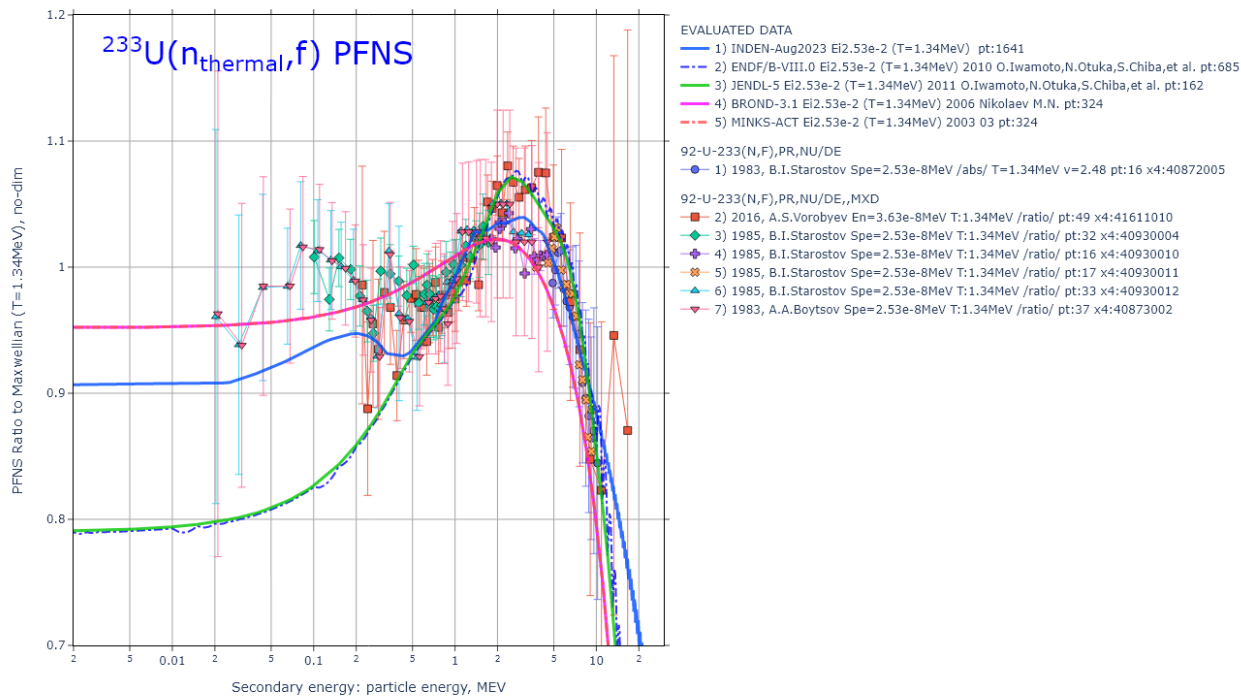
36) part2-B-pfns/out00/pfns233u0_5.html.png

Reaction: 92-U-233(N,F),PR,NU/DE,,REL Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 66
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:03



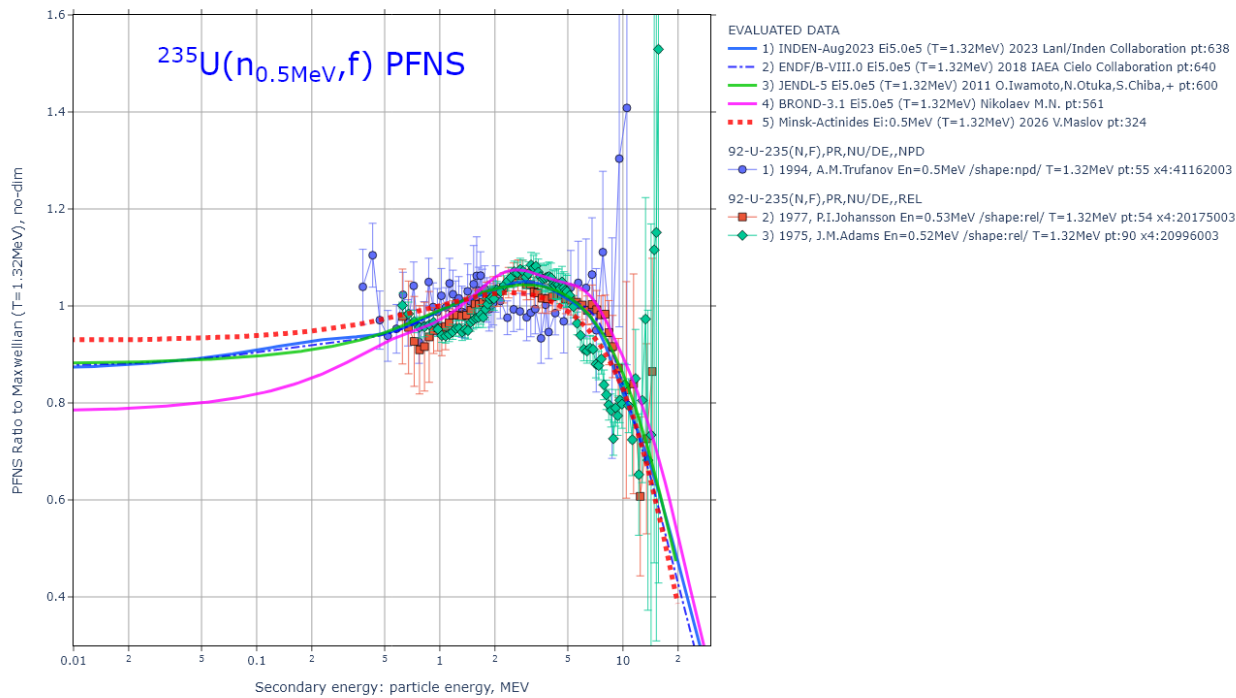
37) part2-B-pfns/out00/pfns233u_t.html.png

Reaction: 92-U-233(N,F),PR,NU/DE Quantity: MFQ:y=Data(Spe,E2) Datasets: 7 Points: 200
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:01



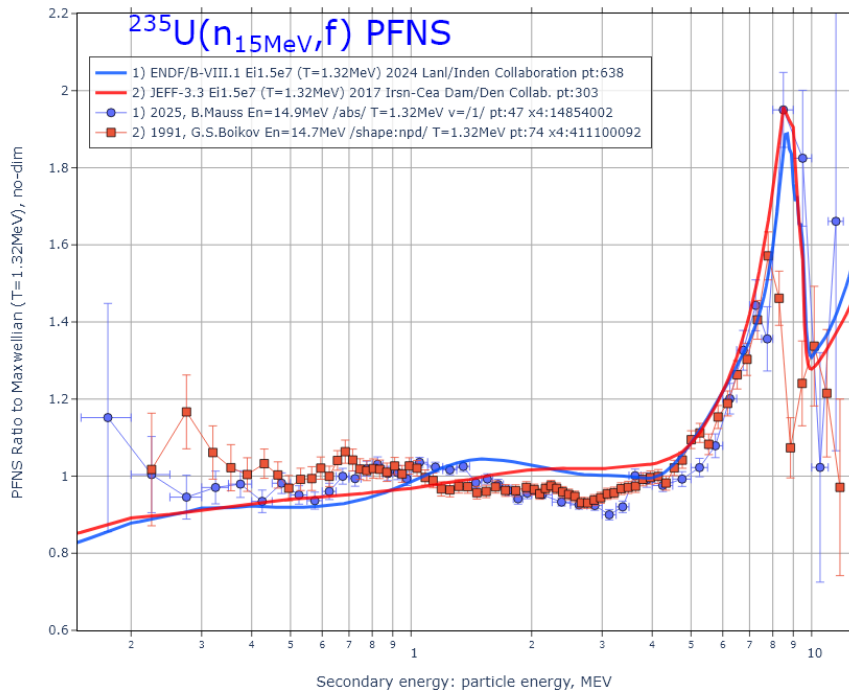
38) part2-B-pfns/out00/pfns235u0_5.html.png

Reaction: 92-U-235(N,F),PR,NU/DE,,NPD Quantity: MFQ:y=Data(En,E2) Datasets: 3 Points: 199
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:07



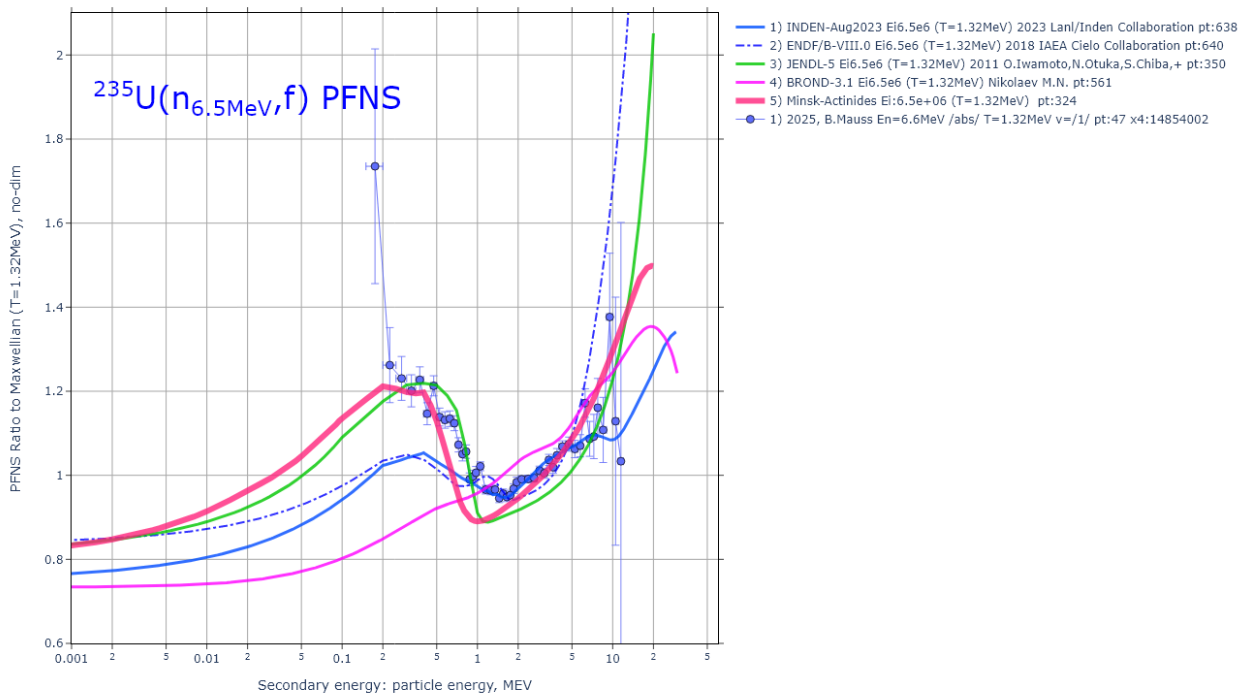
39) part2-B-pfns/out00/pfns235u15.html.png

Reaction:92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:2 Points:121
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-24 //running:2026-09-25 00:45:39



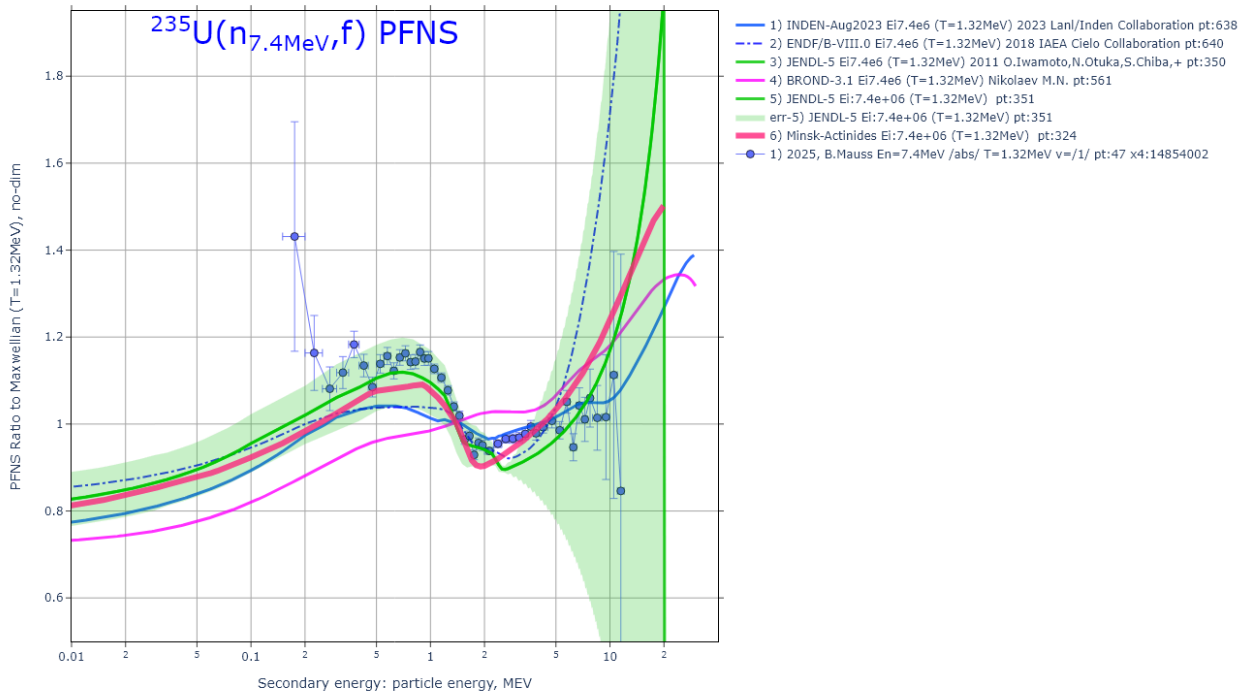
40) part2-B-pfns/out00/pfns235u6_6.html.png

Reaction:92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:1 Points:47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:15



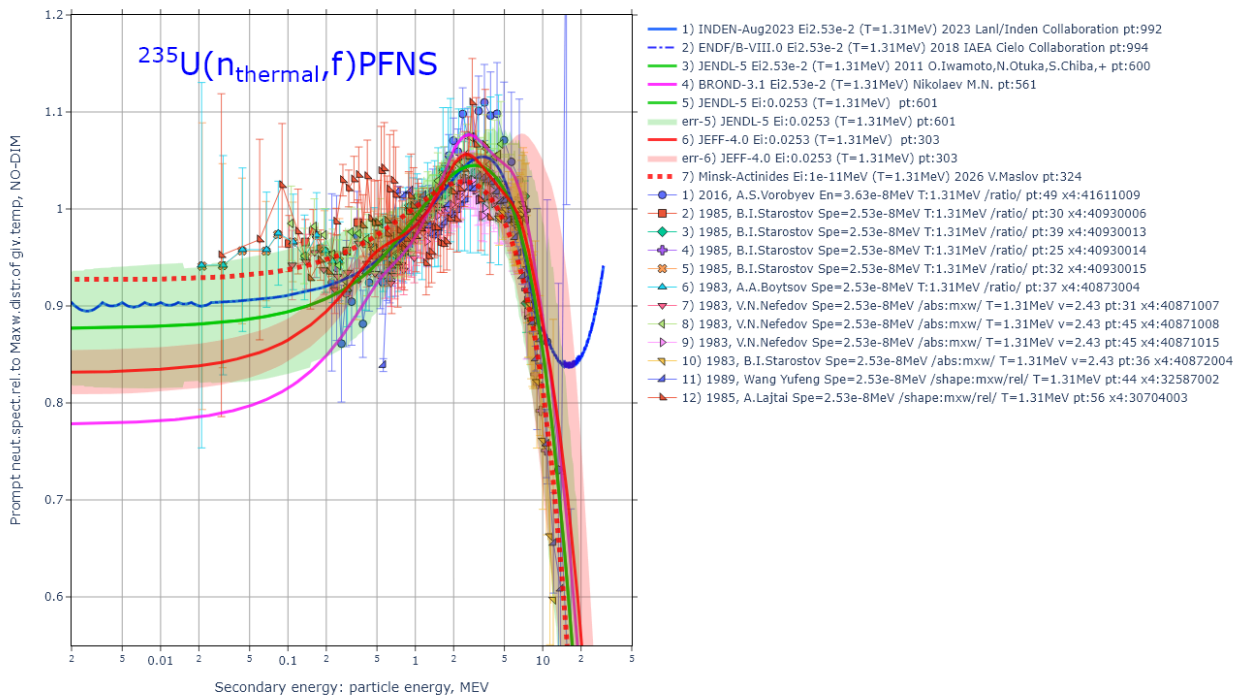
41) part2-B-pfns/out00/pfns235u7_4.html.png

Reaction:92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:1 Points:47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:10

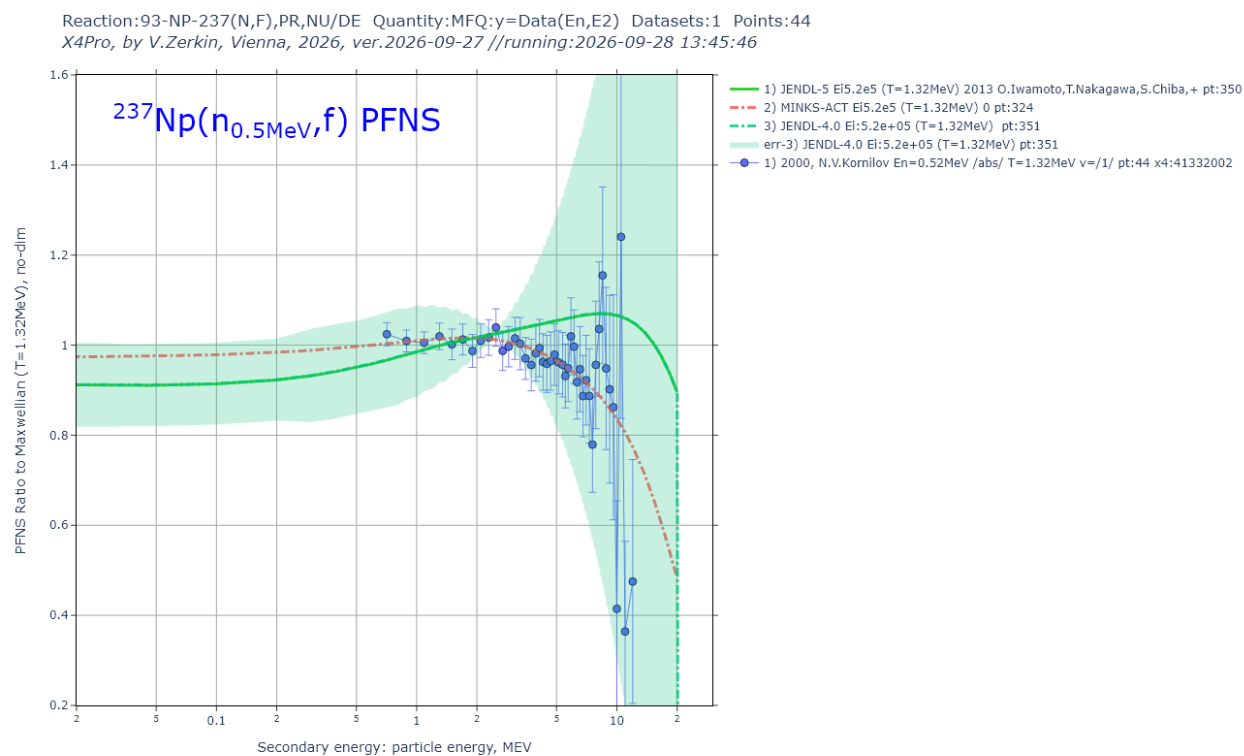


42) part2-B-pfns/out00/pfns235u_t.html.png

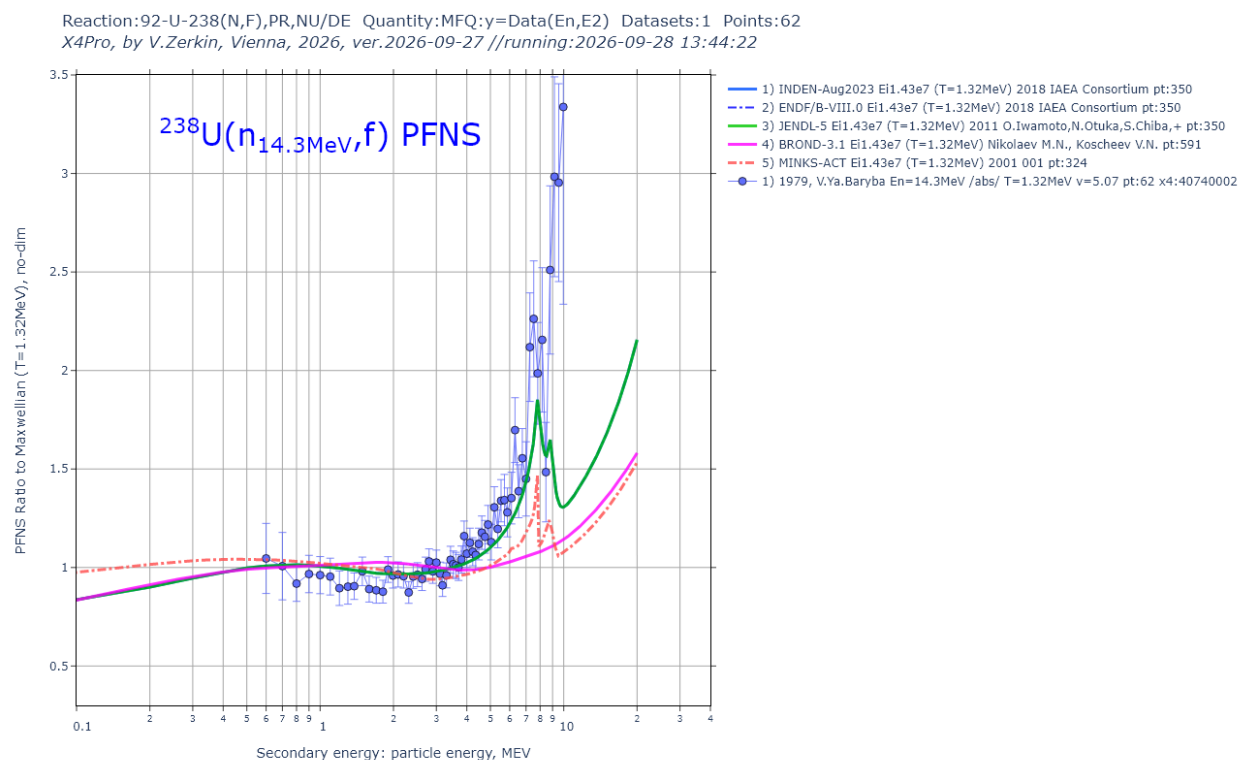
Reaction:92-U-235(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(En,E2) Datasets:12 Points:469
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:05



43) part2-B-pfns/out00/pfns237np0_5.html.png

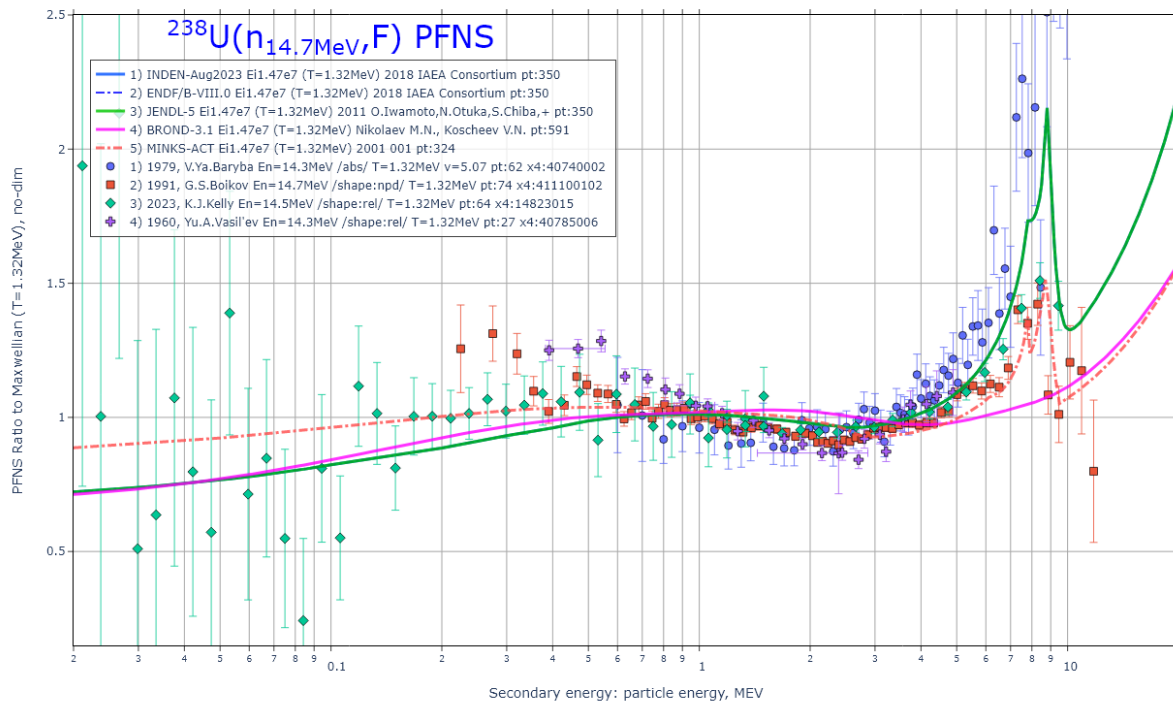


44) part2-B-pfns/out00/pfns238u14_3.html.png



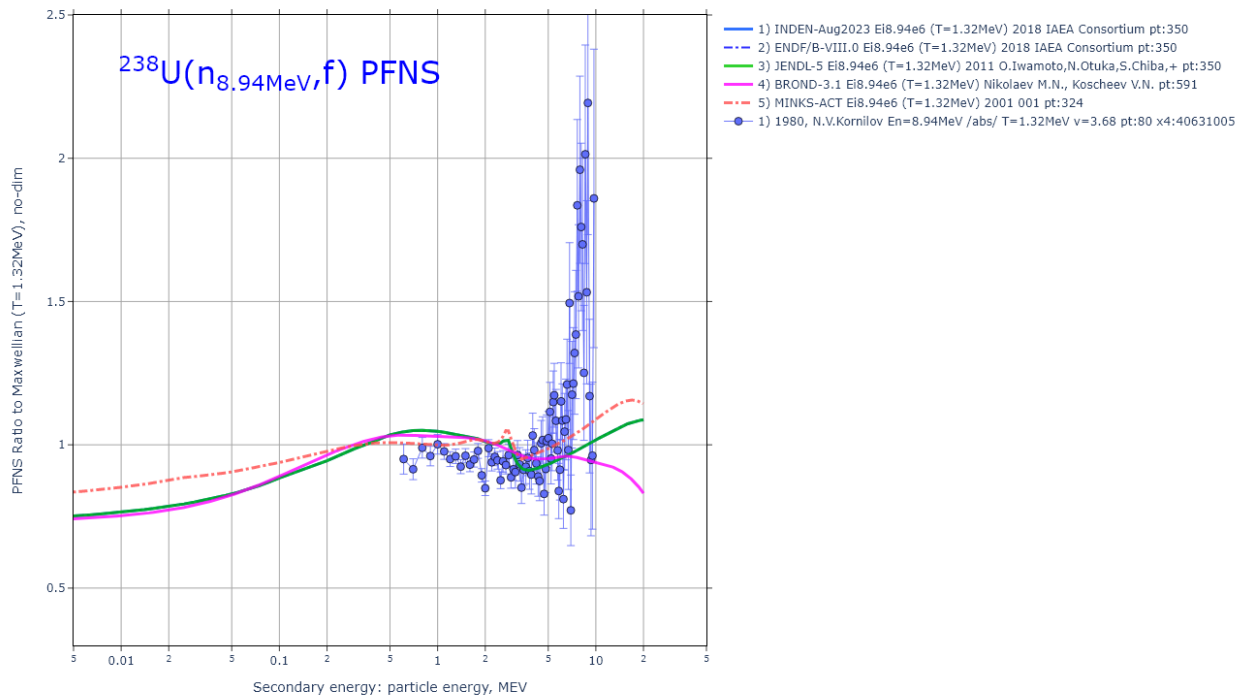
45) part2-B-pfns/out00/pfns238u14_7.html.png

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4 Points: 227
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:25



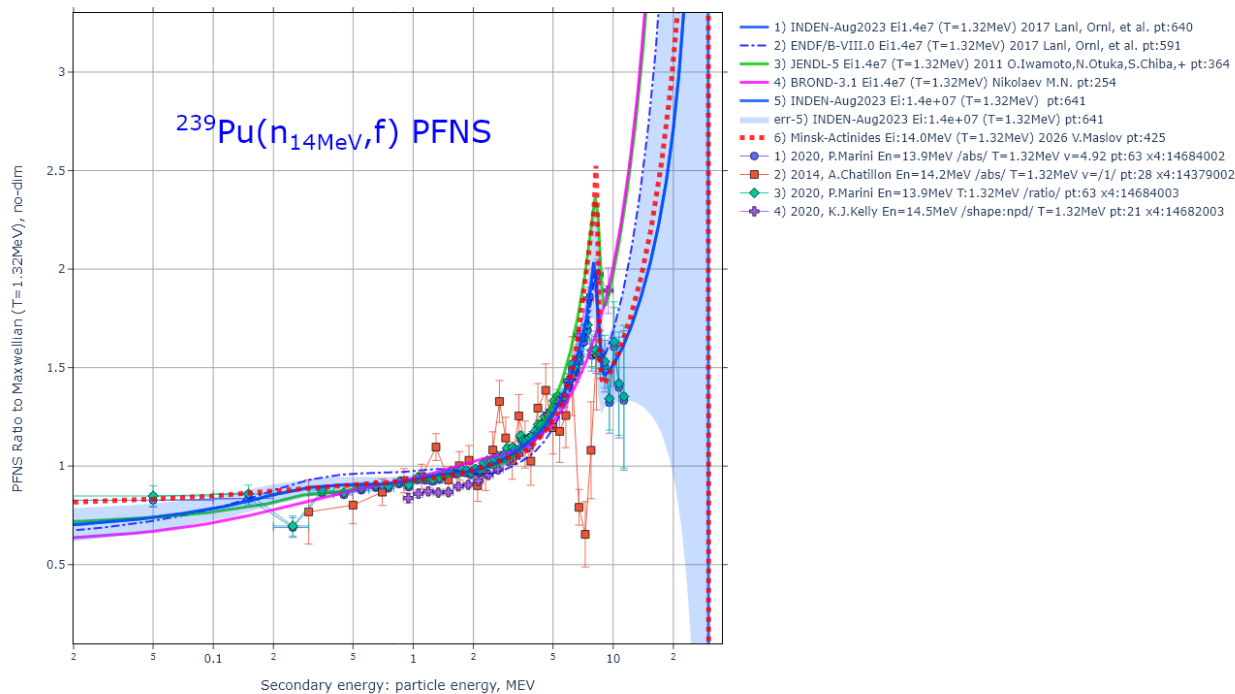
46) part2-B-pfns/out00/pfns238u8_94.html.png

Reaction: 92-U-238(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 1 Points: 80
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:20



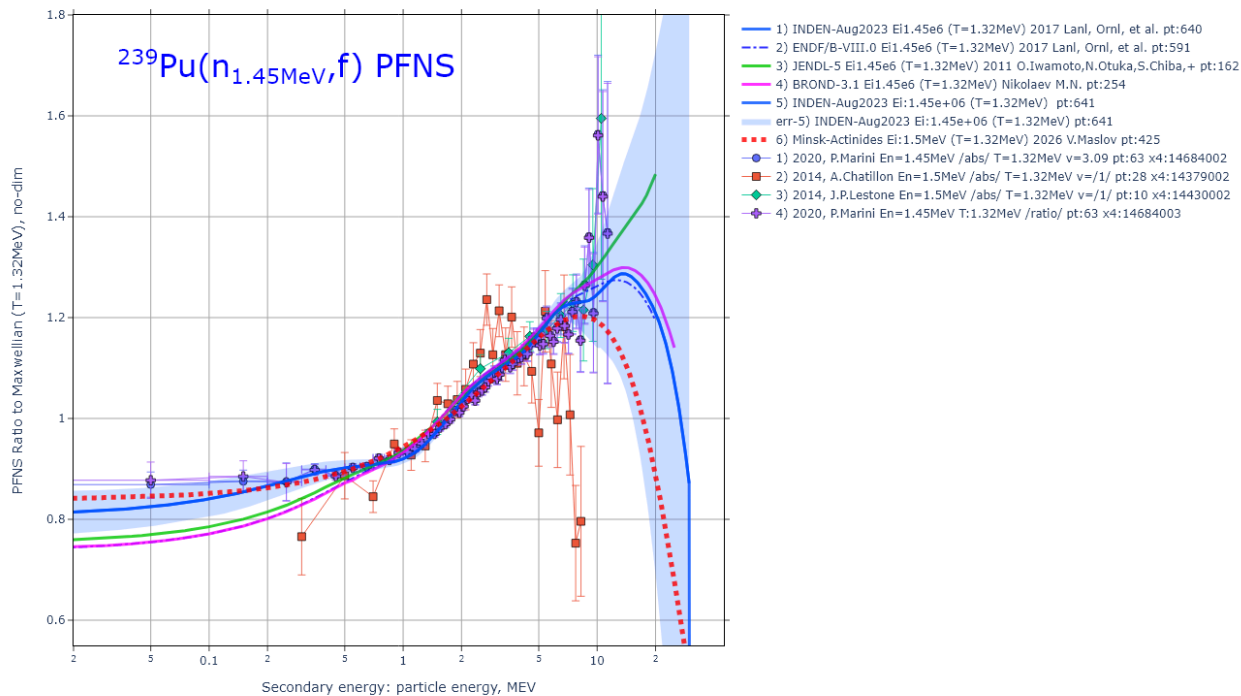
47) part2-B-pfns/out00/pfns239pu14.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4/5 Points: 175/215
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:36



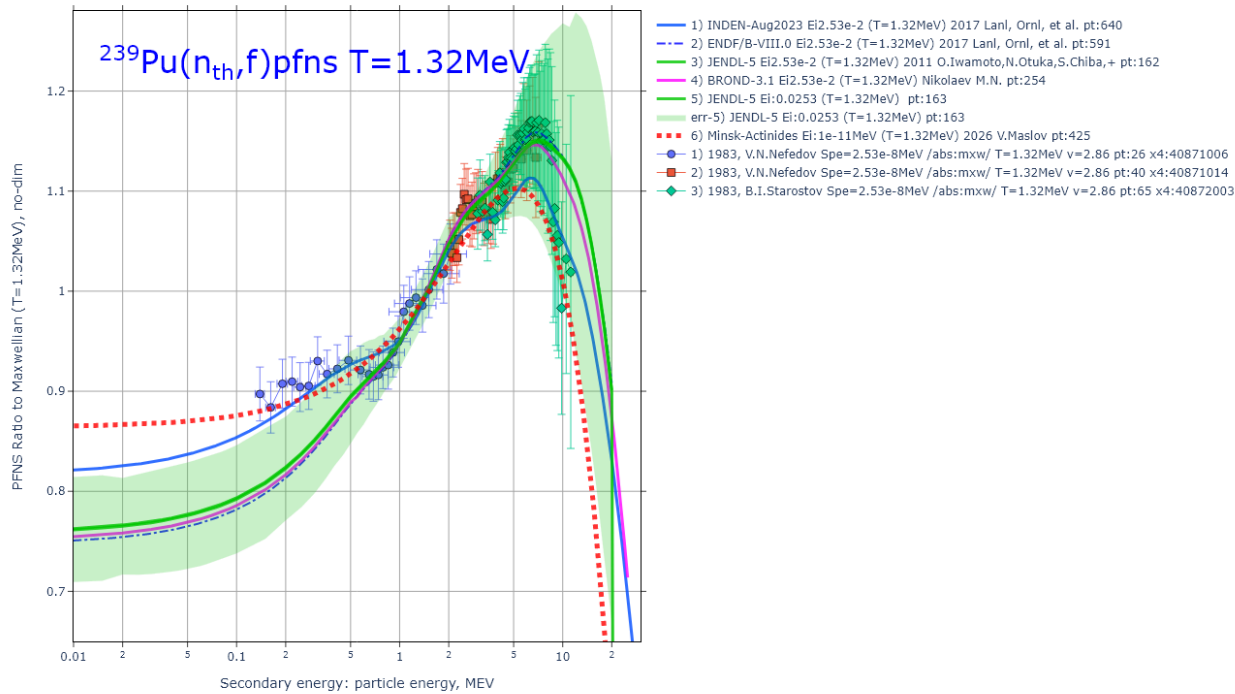
48) part2-B-pfns/out00/pfns239pu1_45.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE Quantity: MFQ: y=Data(En,E2) Datasets: 4 Points: 164/165
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:28



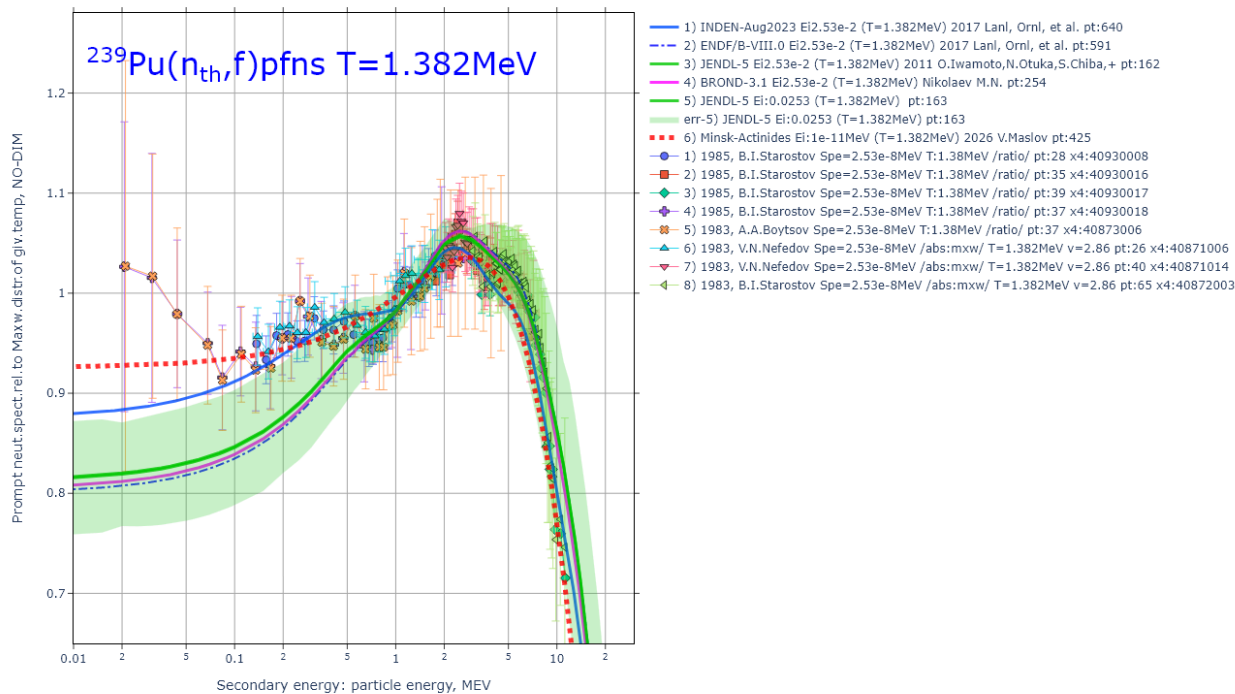
49) part2-B-pfns/out00/pfns239pu_t.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:27



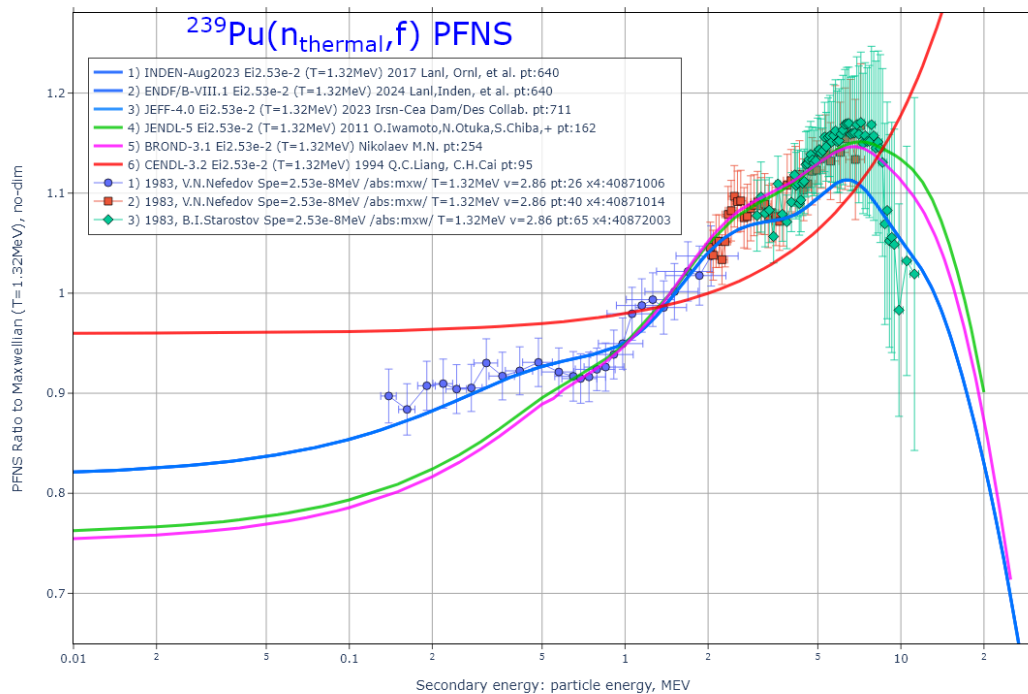
50) part2-B-pfns/out00/pfns239pu_t138.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(Spe,E2) Datasets:8 Points:307
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:44:29



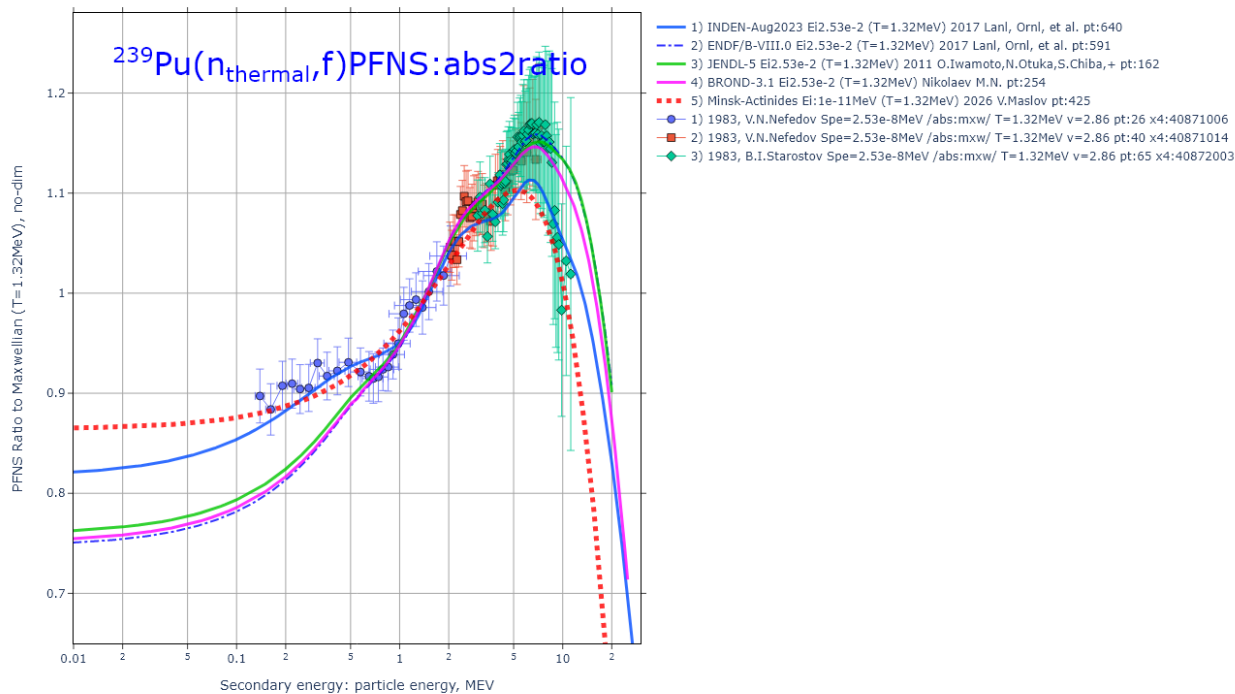
51) part2-B-pfns/out00/pfns239pu_t2.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-20 //running:2026-09-22 19:02:32



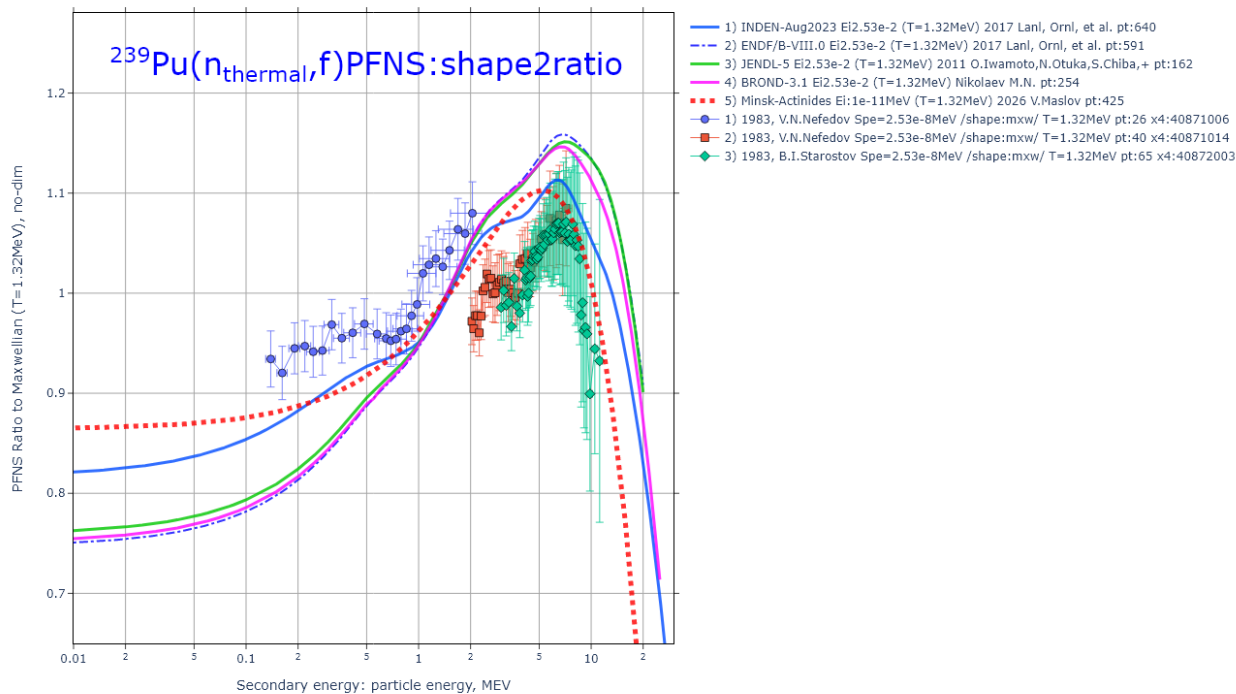
52) part2-B-pfns/out00/pfns239pu_t_abs.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:47



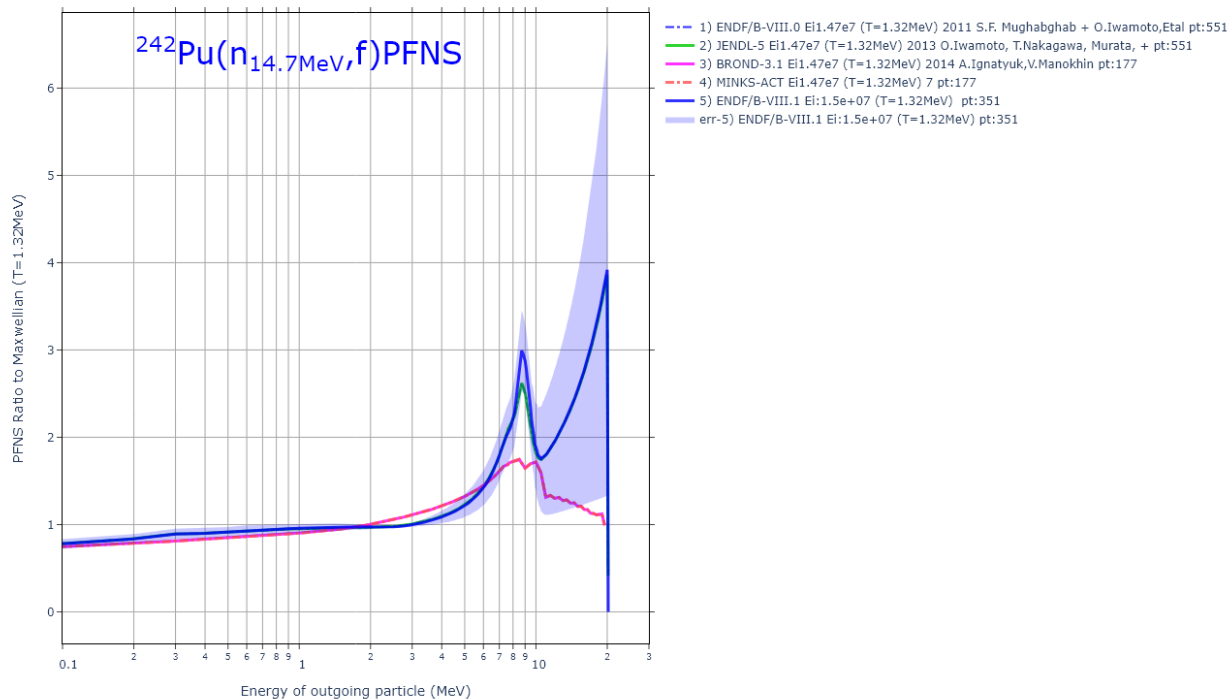
53) part2-B-pfns/out00/pfns239pu_t_shp.html.png

Reaction: 94-Pu-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:49



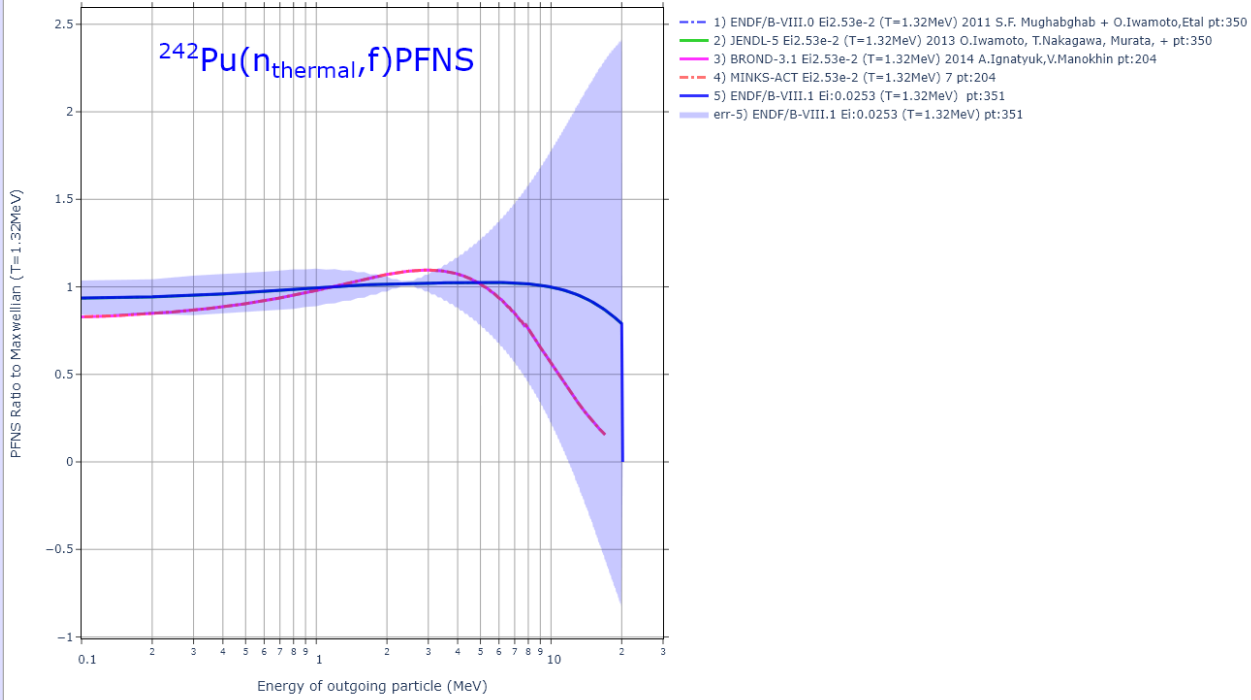
54) part2-B-pfns/out00/pfns242pu14_7.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, Einc=14.7MeV, $T=1.32\text{MeV}$
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:57



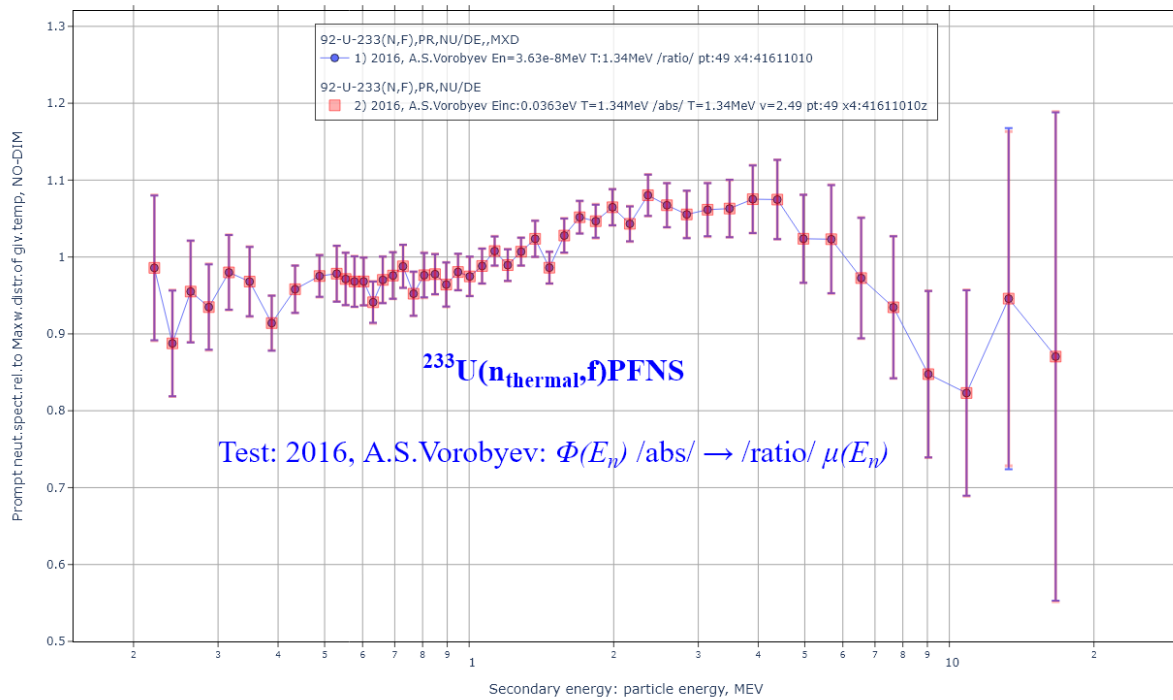
55) part2-B-pfns/out00/pfns242pu_t.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, thermal, T=1.32MeV
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-27 //running:2026-09-28 13:45:53



56) part2-B-pfns/out00/zabs2mxw233u_t.html.png

Reaction: 92-U-233(N,F),PR,NU/DE,,MXD Quantity:MFQ:y=Data(En,E2) Datasets:2/1 Points:98/49
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-20 //running:2026-09-22 17:58:10

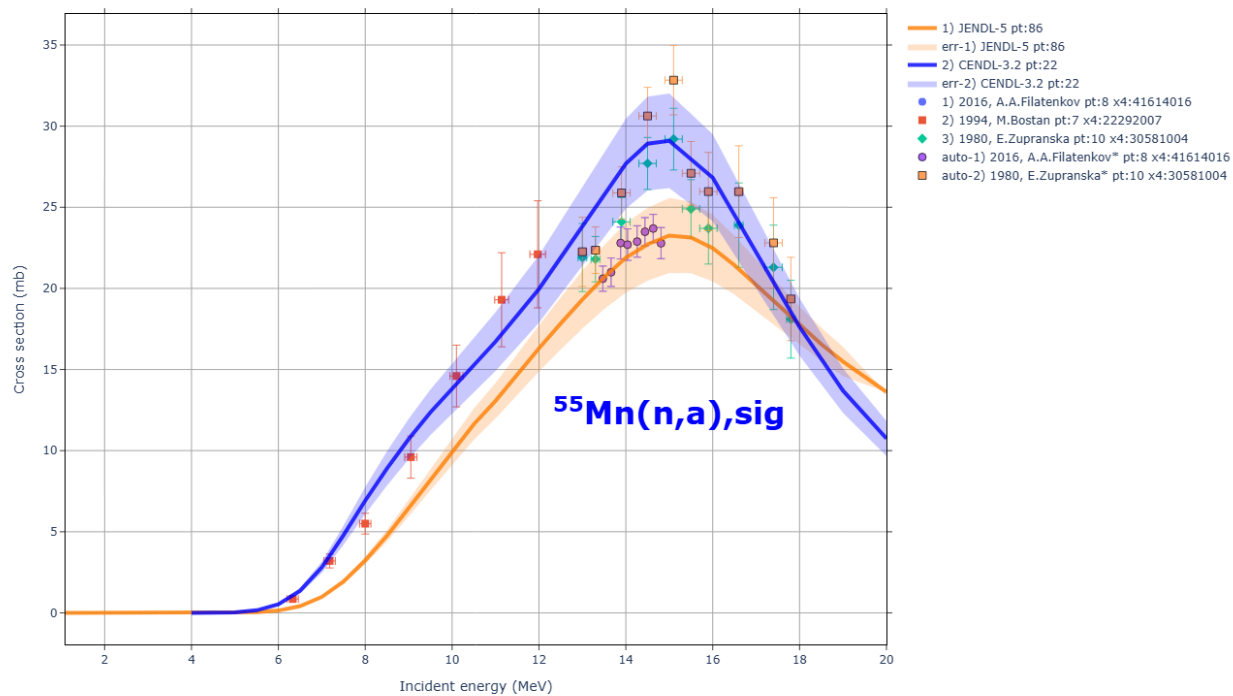


part3-1-auto1/ Automatically renormalize EXFOR CS using old monitor CS and new standards

57) part3-1-auto1/out00/auto1.html.png

Automatic correction of EXFOR cross sections: Mn-55(n,a),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:58:19

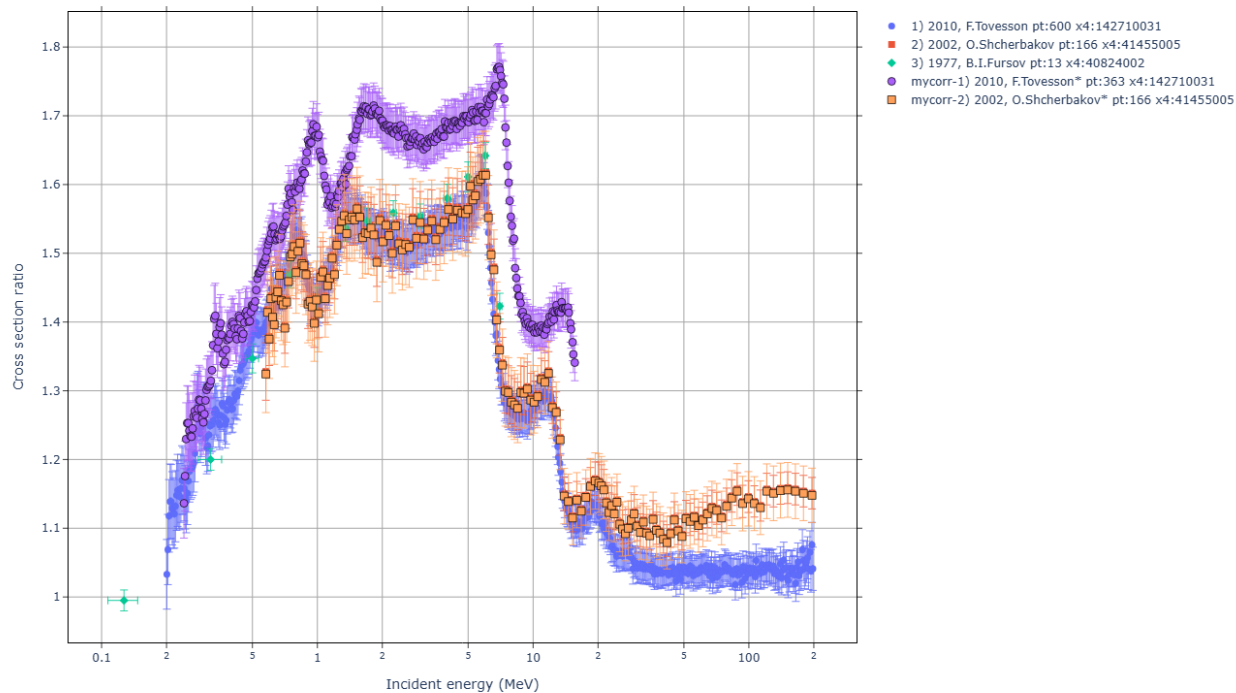


part3-2-user1/ Correct data from EXFOR by user's program

58) part3-2-user1/out00/user1.html.png

Local user's corrections of EXFOR cross sections ratios: Pu-239/U-235(n,f)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:59:05

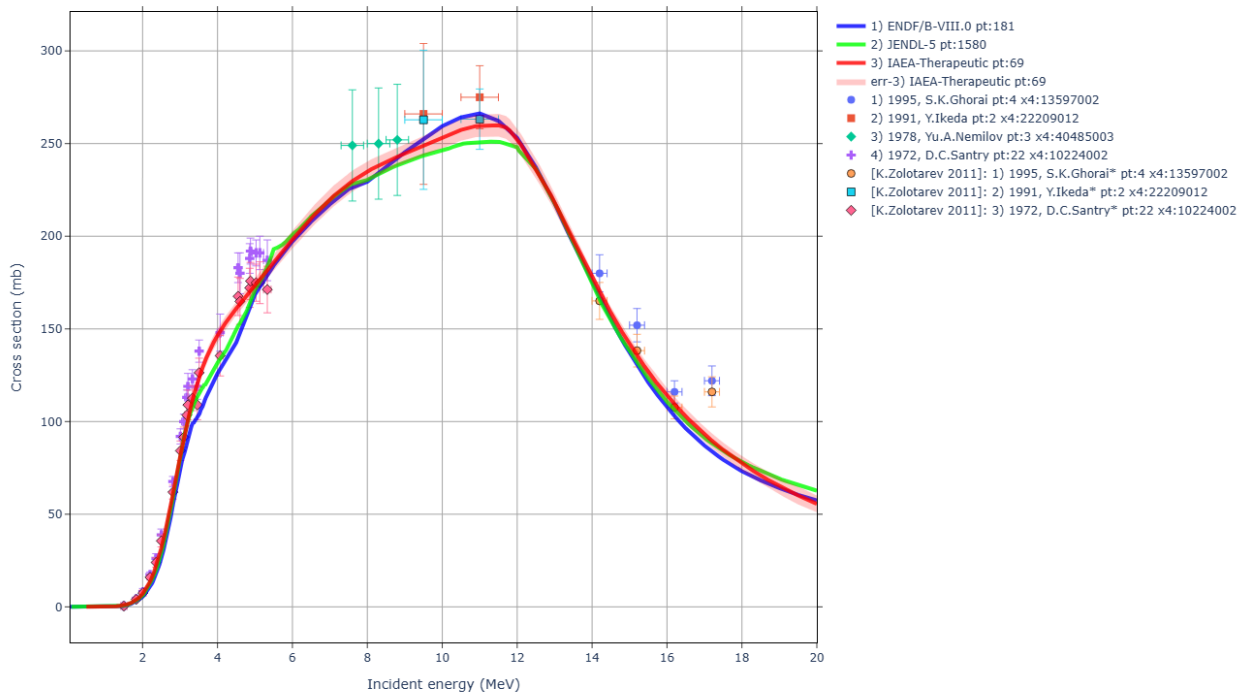


part3-3-expert1/ Apply experts corrections stored in X4Pro database

59) part3-3-expert1/out00/expert1.html.png

Apply experts corrections from database to EXFOR data: Zn-64(n,p),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:59:09

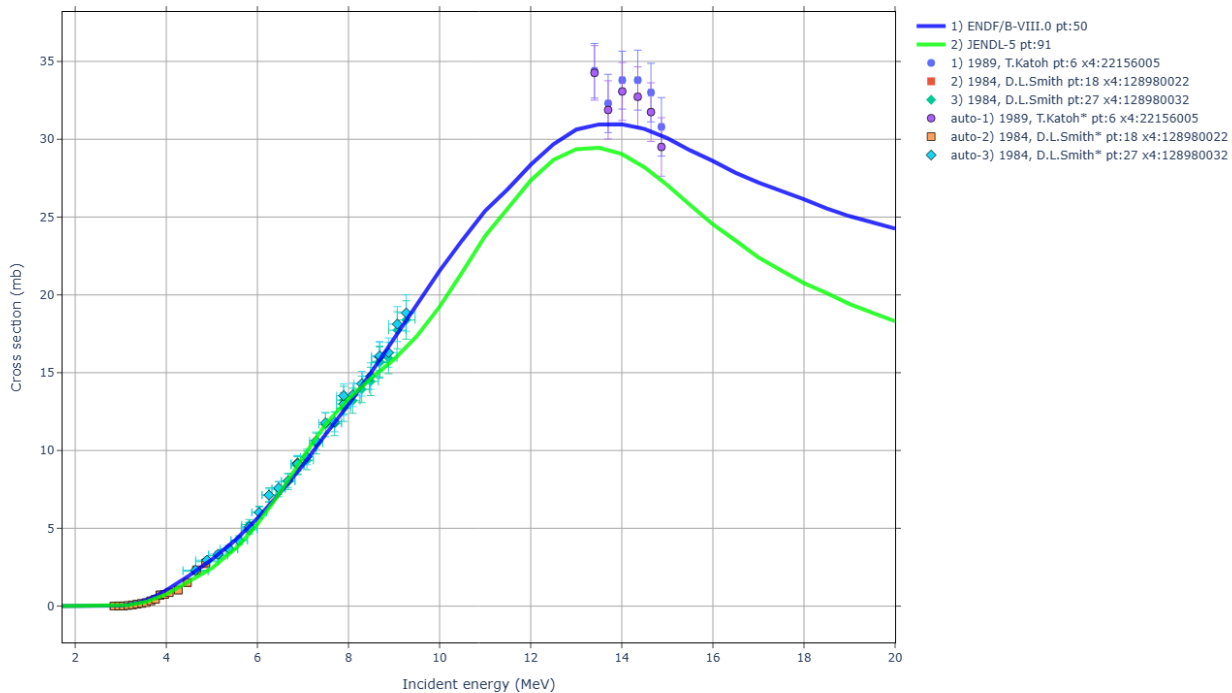


part3-4-auto2/ Automatically renormalize EXFOR CS using old monitor CS and new standards, old and new decay data

60) part3-4-auto2/out00/auto2.html.png

Automatic correction of EXFOR cross sections: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:59:56

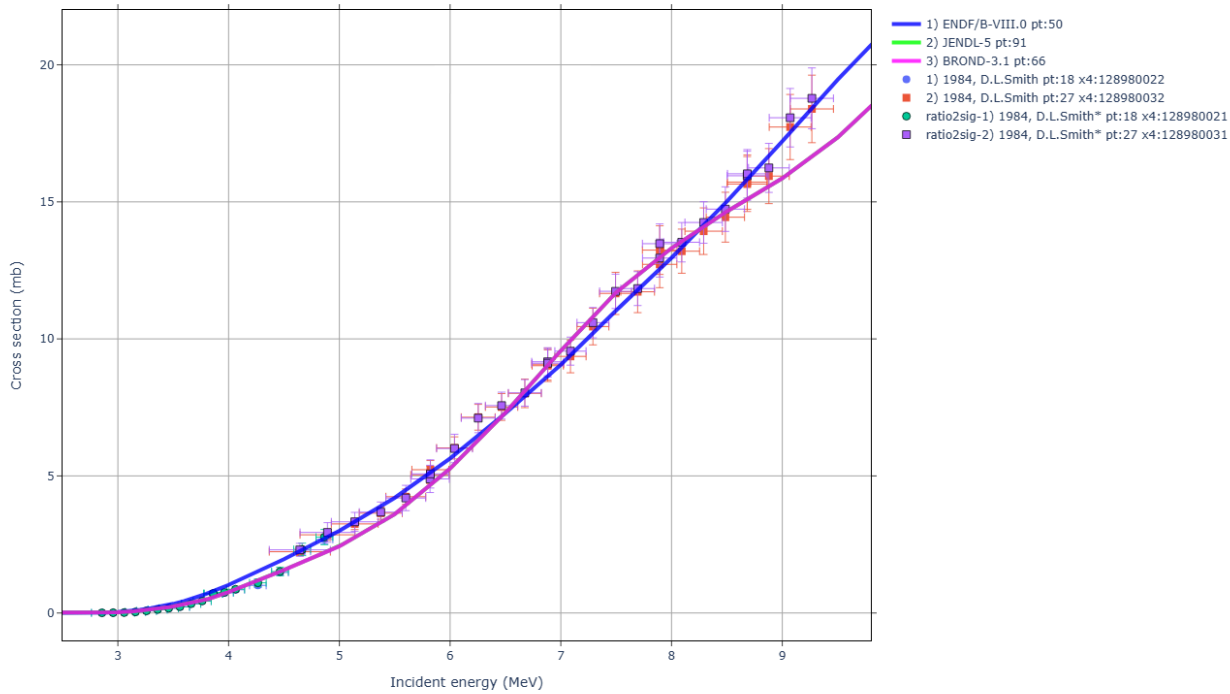


part3-5-ratio2sig/ Retrieve ratios, calculate CS and uncertainties using latest standards

61) part3-5-ratio2sig/out00/ratio2cs.html.png

Ratio to cross section of EXFOR data: V-51(n,p)CS

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 22:00:39

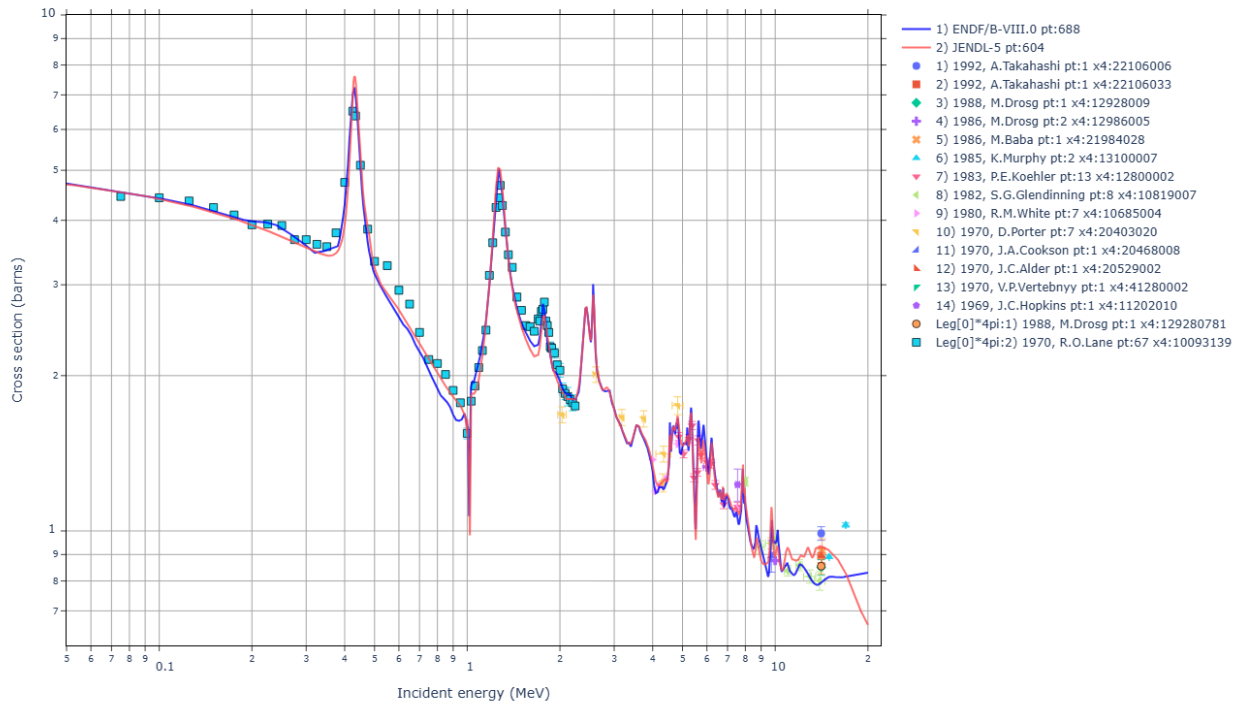


part3-6-daleg2sig/ Retrieve from EXFOR: Legendre coefficient $L[0]$, calculate cross sections and errors

62) part3-6-daleg2sig/out00/B11nel.html.png

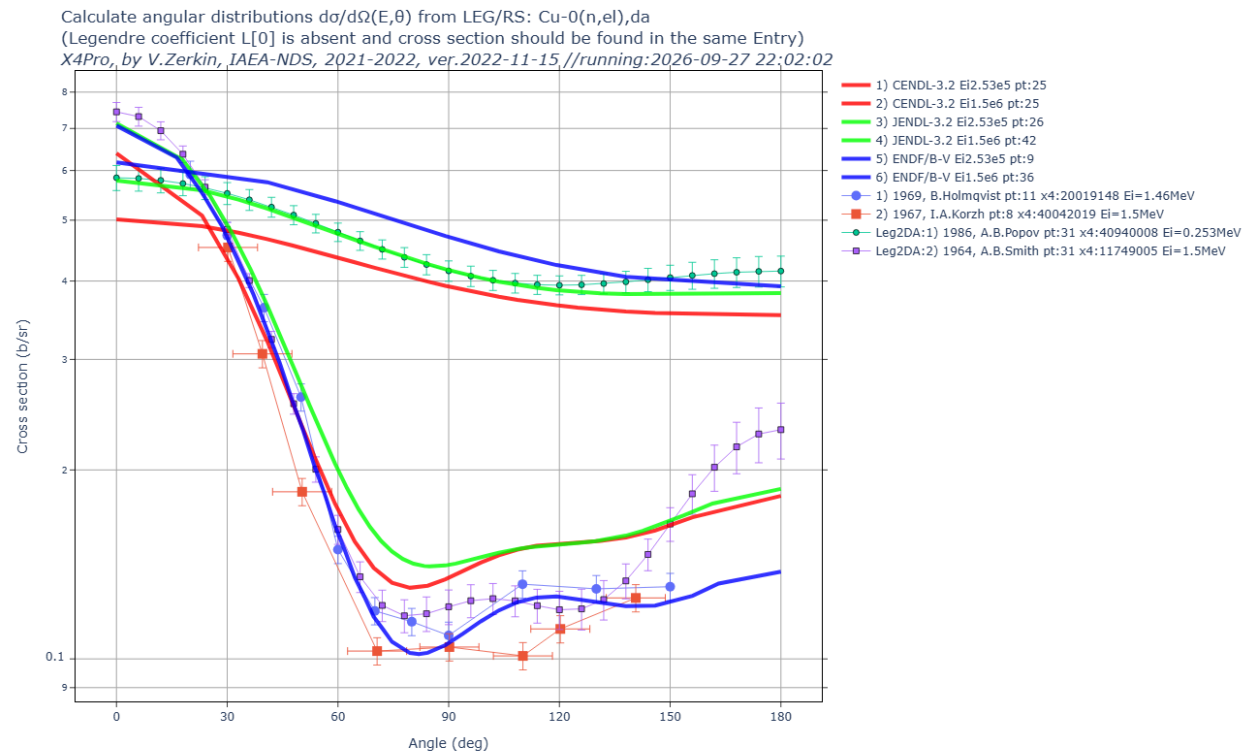
Cross sections $\sigma(E)$: from B-11(n,el),SIG and DA,,LEG:Legendre coeff.L[0]*4pi

X4Pro, by V.Zerkin, IAEA-NDS, 2021-12-07--2022-11-09 //running:2026-09-27 22:01:19



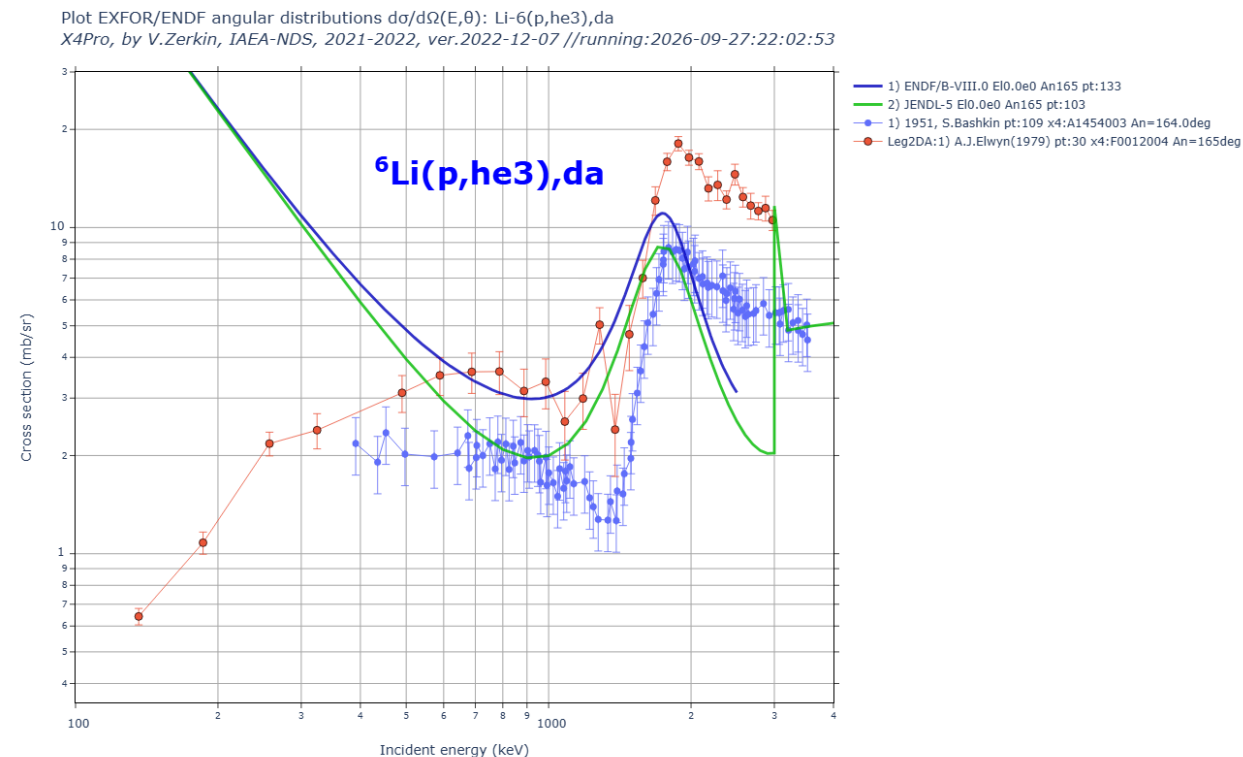
part3-7-legrs2da/ EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre coefficient $L[0]$; calc. angular distributions

63) part3-7-legrs2da/out00/Cu0nel_da.html.png



part3-8-leg2r33/ Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)

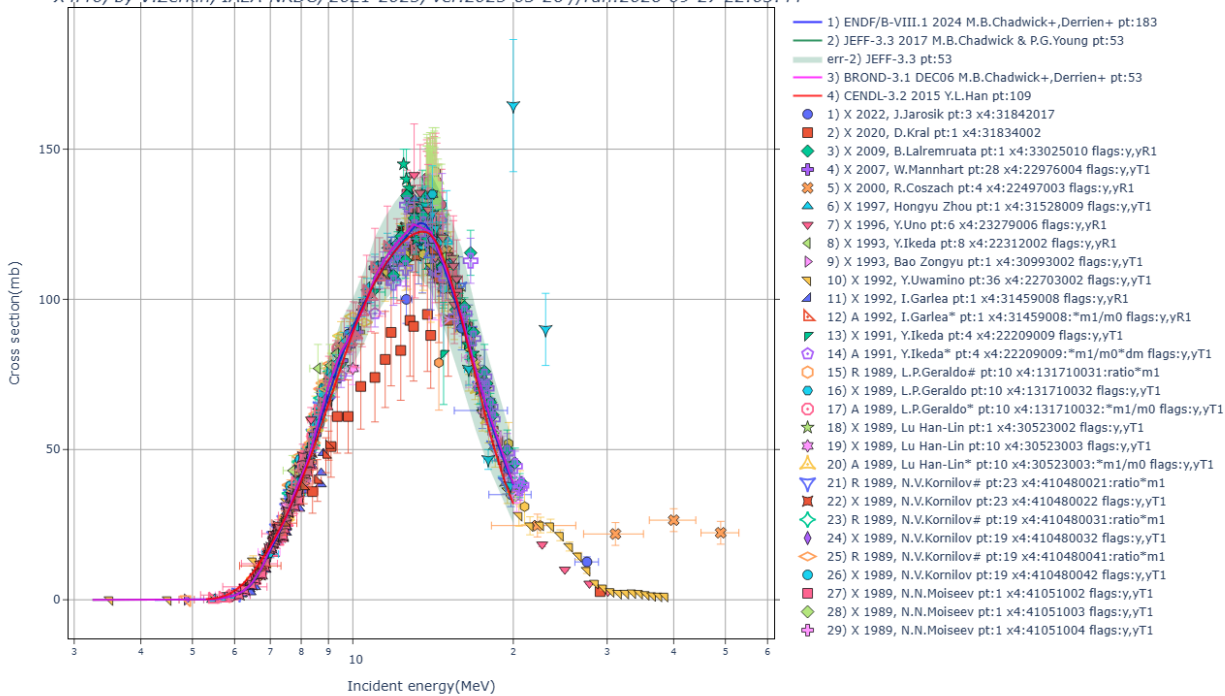
64) part3-8-leg2r33/out00/da1ei_leg2da.html.png



part6-1-eval2sig/ Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS

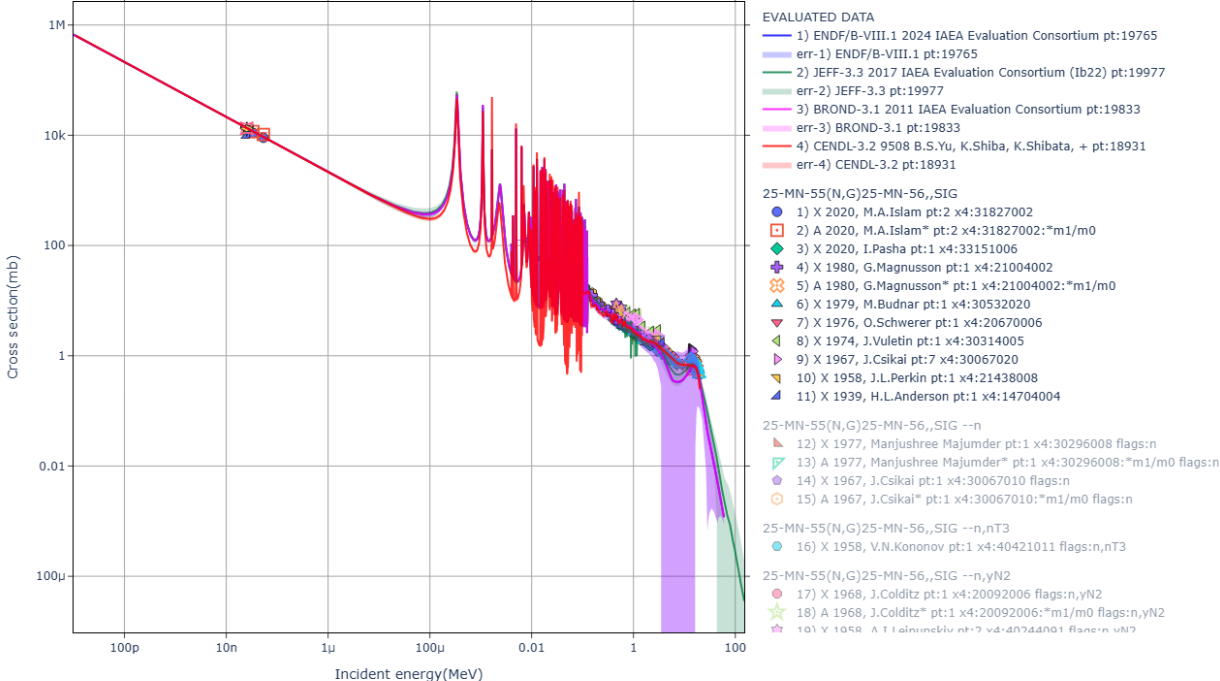
65) part6-1-eval2sig/out00/Al27na.html.png

Cross sections $\sigma(E)$: Al-27(n,a),sig datasets:116 original:85X auto-corrected:22A* ratio2cs:9R#
Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-09-27 22:03:44

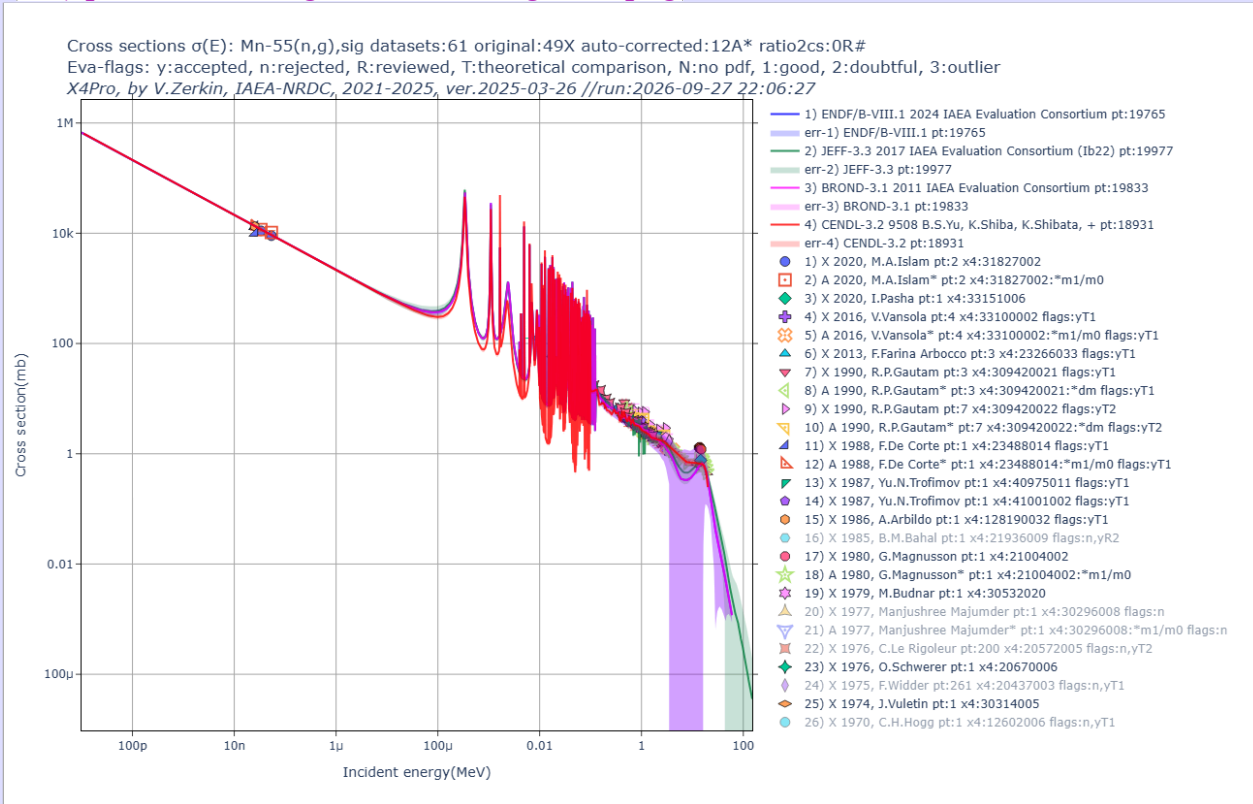


66) part6-1-eval2sig/out00/Mn55ng-n.html.png

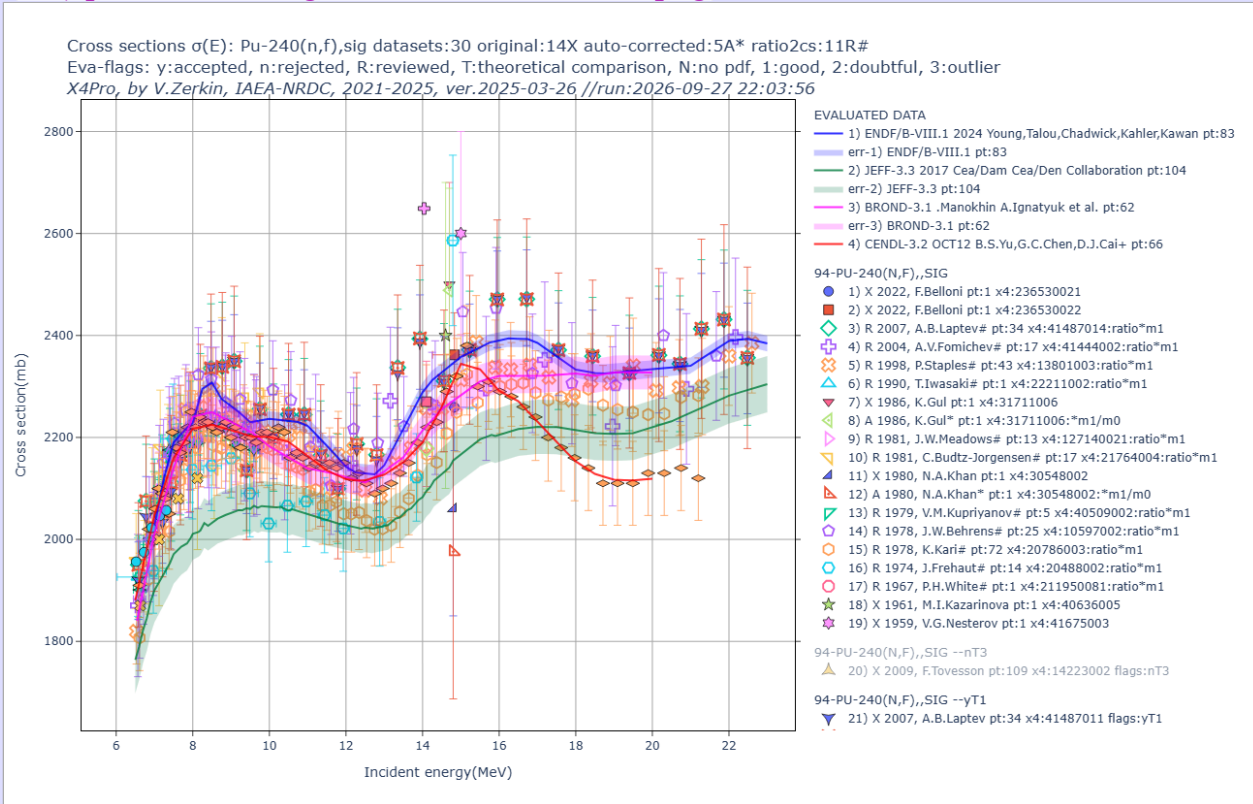
Cross sections $\sigma(E)$: Mn-55(n,g),sig datasets:61 original:49X auto-corrected:12A* ratio2cs:0R#
Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-09-27 22:03:48



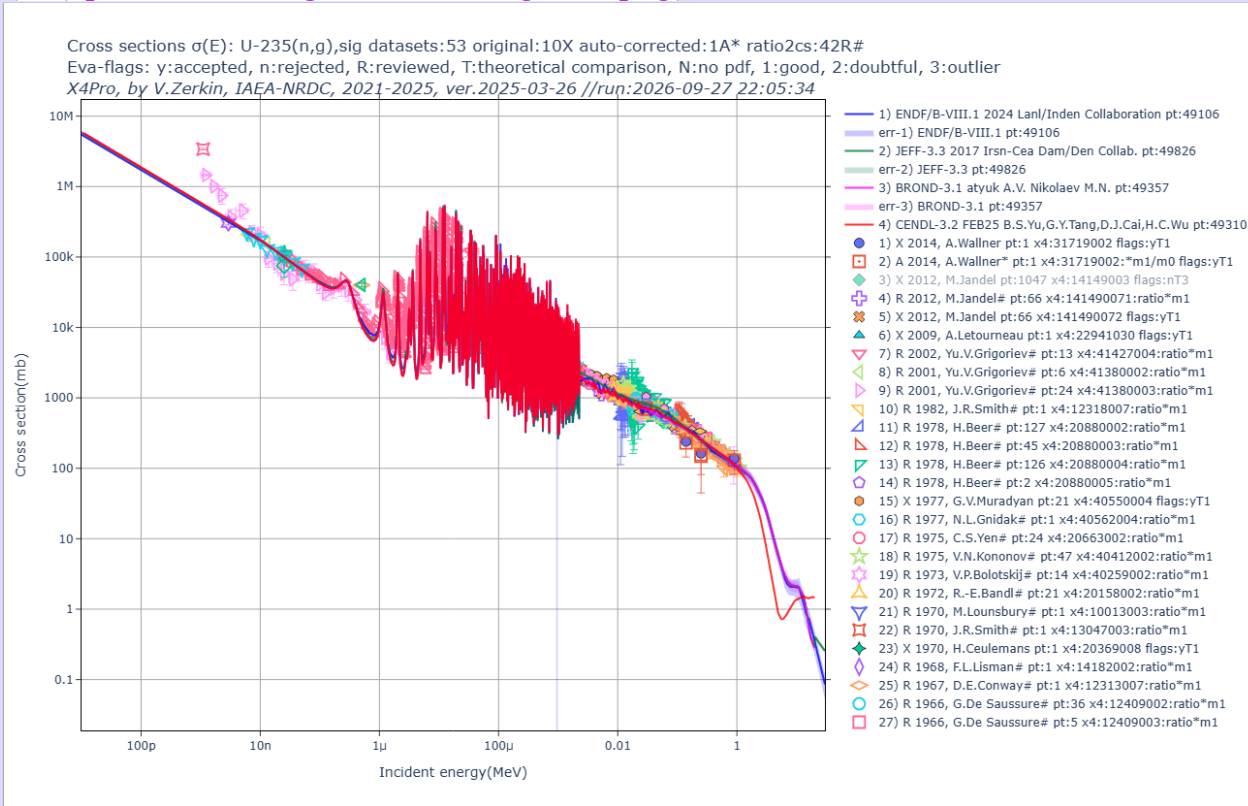
67) part6-1-eval2sig/out00/Mn55ng.html.png



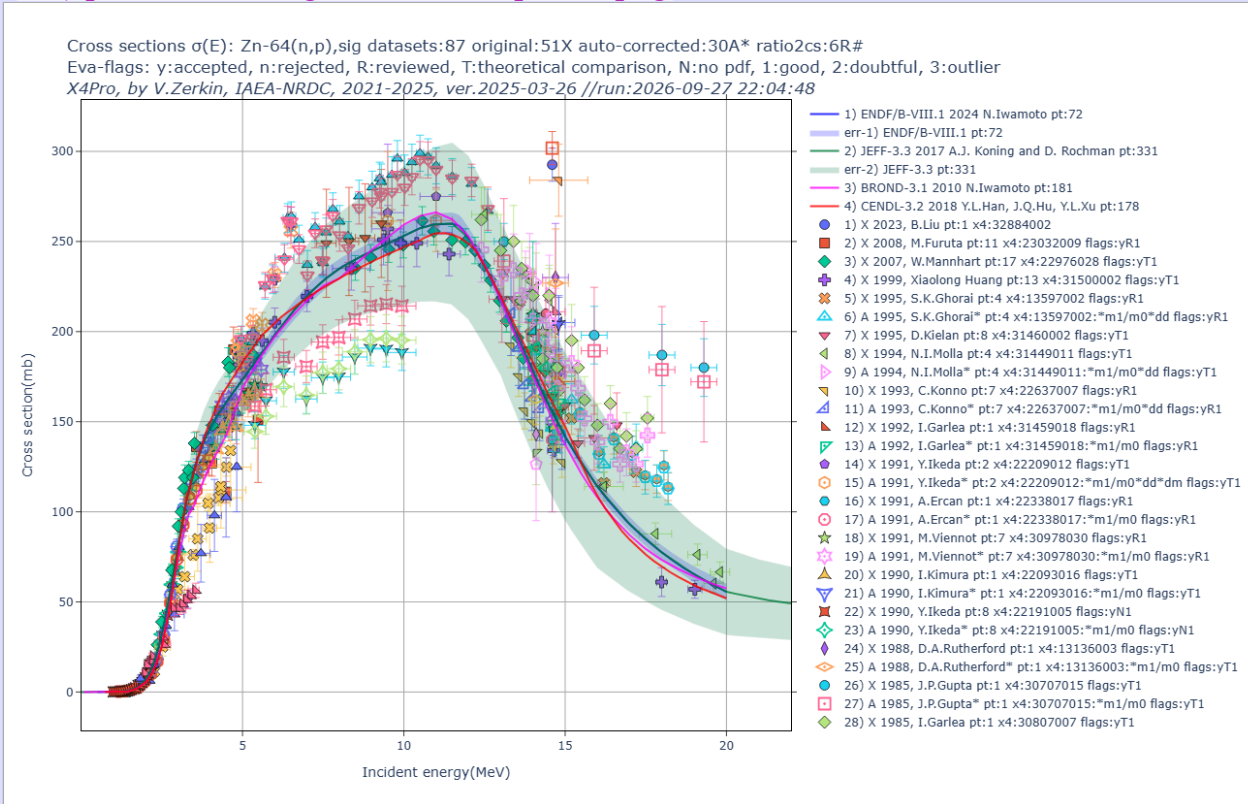
68) part6-1-eval2sig/out00/Pu240nf-n.html.png



69) part6-1-eval2sig/out00/U235ng.html.png



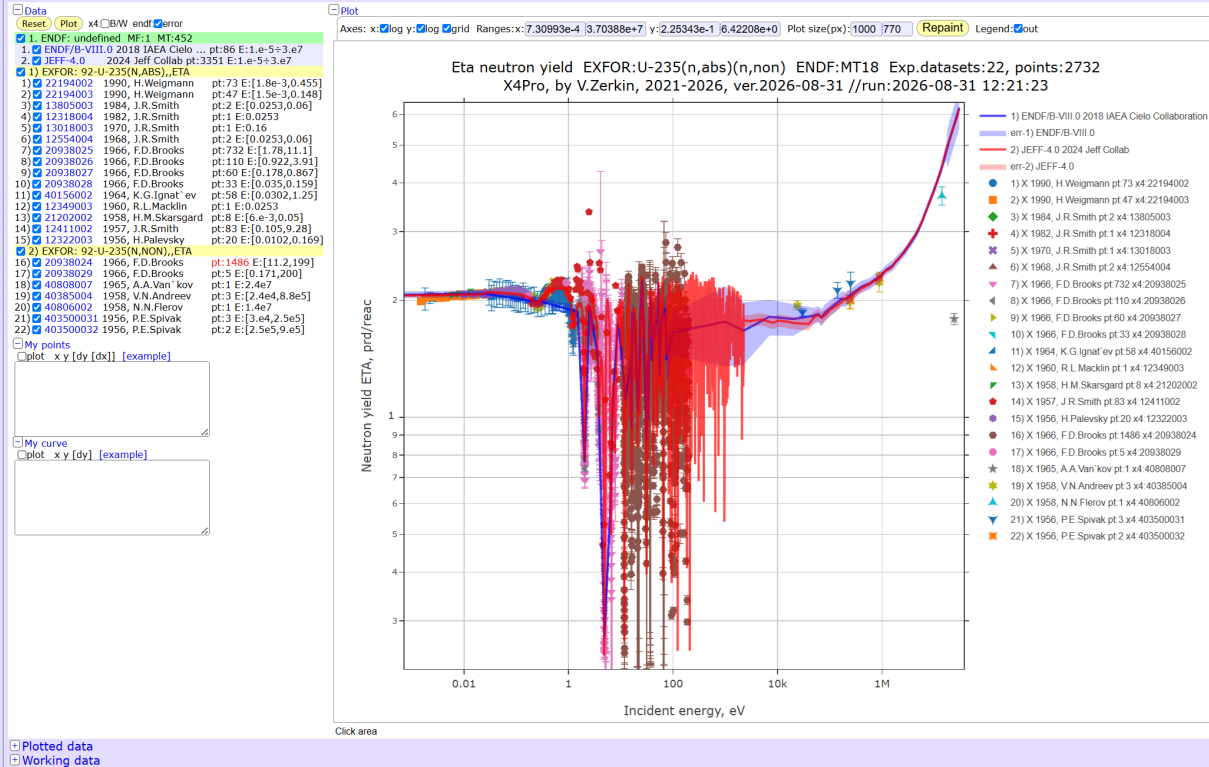
70) part6-1-eval2sig/out00/Zn64np.html.png



71) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT452,18,102,MF1,3,31,33

X4Pro - advanced plot

by V.Zerkin, 2025-2026



Created by V.Zerkin, 09-Aug-2026
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